

DYNAMICAL SHAKE-UP OF PLANETARY SYSTEMS. I. EMBRYO TRAPPING AND INDUCED COLLISIONS BY THE SWEEPING SECULAR RESONANCE AND EMBRYO-DISK TIDAL INTERACTION

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ABSTRACT

We analyze the orbital evolution of terrestrial planetary embryos after oligarchic growth, including the effect of the sweeping Jupiter secular resonance combined with tidal drag during the depletion of the protoplanetary gas disk. Previous studies show that the orbits of isolated embryos become unstable through long-term gravitational interaction. However, the planetary systems formed as a result of giant impacts are generally very eccentric, unlike our solar system. Although mechanisms that damp the eccentricity have been proposed, such mechanisms not only damp the eccentricity of embryos strongly but also prevent their orbital crossing and planetary growth. This dilemma can be solved if the protoplanetary collisions occur in an environment where the damping is still working. **We consider the stage of gas depletion after the formation of a Jupiter-like planet in the disk. The gas depletion changes the gravitational potential and causes the sweeping of a secular resonance.** We find that the secular resonance passes through the terrestrial region from outside to inside, that the resonance excites the eccentricities of isolated embryos, and that it leads to orbit crossing. Because the remnant disk is still present in this stage, the gas drag due to tidal interaction is still effective. The tidal drag effectively damps the eccentricities of the fully grown embryos. This process allows the system to form circular orbits analogous to our solar system. We also find that the tidal drag induces the decay of the semimajor axes of the embryos. As a result of balance between damping and excitation of eccentricity, the embryos migrate along with the secular resonance. This “secular resonance trapping” can lead to rapid collisions and mergers among the embryos as they migrate from the outer to the inner region, concurrent with the disk depletion. Since the induced migration is strongest near the Jovian orbit, the final terrestrial planets tend to be concentrated in a relatively small region (< 2 AU). We suggest that this mechanism may be the origin of the severe present-day mass depletion of the asteroid belt.

Subject headings: celestial mechanics — planetary systems: formation — solar system: formation

1. INTRODUCTION

The major planets in the solar system are naturally divided into three groups: four terrestrial planets within 2 AU, two intermediate-period gas planets, and two distant ice giant planets. Other than Mercury, the orbits of these planets all have relatively low eccentricities, lending them dynamical stability over at least the age of the solar system. Around other stars, planetary systems may (1) attain protracted maintenance of nearly circular orbits, (2) undergo modest eccentricity modulation, or (3) become dynamically unstable. These evolutionary outcomes have direct implications on the physical environment of potentially habitable planets. In the quest to address the key astrobiological issue, “are we alone in the universe?”, it is particularly relevant to examine whether the dynamical architecture of the solar system is unique.

In principle, this issue must be ultimately resolved by observations of extrasolar planetary systems. Over 100 Jupiter-mass planets have been discovered around nearby stars with precise radial velocity measurements (for a review see Marcy et al. 2003). In the foreseeable future, the range of periods and amplitude sensitivity accessible to these surveys are expected to increase to longer periods and higher precisions (e.g., Carter et al. 2003;

Fischer et al. 2003; Jones et al. 2003). In addition, several programs have been developed in attempts to directly search for terrestrial planets through transit observations, astrometrical measurements, and gravitational lensing events. Future space missions using accurate photometry (the *COROT* and *Kepler* missions), precision astrometry (the *SIM* mission), and high-resolution imaging (the *TPF* and *Darwin* missions) promise to provide the first proof of the existence of terrestrial planets around main-sequence stars other than the Sun. However, an essential component of the engineering design and the optimum observational strategy of these space missions is a reliable assessment of the anticipated frequency of Earth-like planets around Sun-like stars.

In light of this observational quagmire, a fruitful approach to infer the frequency of terrestrial planets around other stars is to carry out a detailed theoretical case study of their formation and dynamical evolution in the solar system. In this context, we are constrained by a rich bank of detailed observational information, ranging from meteoritic records and planetary compositions to the planets’ accurately determined orbital parameters. It is likely that most of these data provide clues on the consequence rather than the causes associated with the formation of terrestrial planets. Some information may reflect the special circumstance under which planets in the solar system were assembled. Our objectives are (1) to determine the causal relations between the planets’ formation processes and their asymptotic dynamical characteristics and (2) to identify some common features and divergent properties of the solar system and planetary systems around other stars.

In this pursuit, we adopt the conventional paradigm of the sequential accretion scenario (SAS; Wetherill 1980; Pollack et al.

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1996). For a fiducial initial condition, we adopt the minimum-mass nebula model (MMNM; Hayashi et al. 1985), which is based on a phenomenological interpretation of the present distribution of heavy elements in the solar system and the assumption that planets have always resided in the vicinity of their birth. The MMNM appears to be consistent with the estimates of heavy-element content in protostellar disks, inferred from the millimeter observations of young stellar objects in the Tauri and ρ Oph clouds (Beckwith & Sargent 1991).

The SAS can be recapitulated in the following three stages: (1) Planetesimals rapidly grow through cohesive collisions into protoplanetary embryos that have sufficient mass to perturb their neighboring planetesimals. In the presence of gas, the embryos attain nearly circular orbits and their growth is stalled by dynamical isolation with several lunar masses (Kokubo & Ida 2002). (2) Beyond the snow line, the embryos' asymptotic mass becomes greater than a few Earth masses. They undergo efficient gas accretion and evolve into Jupiter-mass gas giant planets. The first planet to complete its growth is preferentially located near the snow line (Pollack et al. 1996; Ida & Lin 2004). (3) Gas is depleted in the disk on $\tau_h \sim 3\text{--}10$ Myr timescales (Haisch et al. 2001; Hollenbach et al. 2000; Carpenter et al. 2005) and residual embryos collide to form prototerrestrial planets interior to the snow line and ice giants exterior to the gas giants.

In this paper we focus our attention on the final stage of terrestrial planet formation. The key issue we wish to address is, how did the four terrestrial planets complete their final assemblage in less than 30–50 Myr with nearly circular orbits? The agglomeration of planets requires orbit crossing of their progenitors and a transition from orderly isolation to dynamical chaos. Yet, the planets' present-day circular orbits provide the necessary condition for the preservation of stability on timescales (4.6 Gyr) at least 2 orders of magnitude longer than that of their formation epoch. The long-term stability of the solar system is also an essential assumption for the MMNM. Since extrasolar terrestrial planets have not been discovered yet, it is premature to contemplate whether this issue may be unique to the architecture of the solar system. However, we intend to identify the necessary conditions that enable the emergence of stable planetary configurations around other stars.

There have been several attempts to address this issue. In a pioneering work, Safronov (1969) realized that in a gas-free environment, gravitational interactions between planetesimals excite their velocity dispersion, which enables their orbits to cross (see also Wetherill 1980; Palmer et al. 1993) and grow beyond their isolation masses. Subsequent physical collisions lead to the emergence of a population of protoplanetary embryos (Kokubo & Ida 1998). Dynamical instabilities lead to further excitation of eccentricity among more massive protoplanetary embryos. Although cohesive collisions in a dynamically hot system of protostellar embryos lead to the assemblage of Earth-mass objects on a timescale of ~ 100 Myr, this process also introduces large eccentricities among the resultant protoplanets (Wetherill 1990). While orbit crossing is needed for protoplanets to acquire their masses, it must paradoxically be avoided in the subsequent 4.6 Gyr in order to preserve the orderly structure of the solar system. Another major problem is that, in a minimum-mass nebula, the growth timescales for the merging Earth-mass embryos exceed 50–100 Myr, which appears to be inconsistent with the 30–50 Myr formation timescale inferred for the Earth (Yin et al. 2002).

One potential mechanism for limiting the terrestrial planets' asymptotic eccentricity is dynamical interactions with a background of lower mass embryos (Chambers & Wetherill 1998)

or a residual population of small particles (Goldreich et al. 2004). Through dynamical friction, the kinetic energy associated with the random motion of the protoplanetary embryos is transferred (Stewart & Wetherill 1988) to the background low-mass planetesimals (Aarseth et al. 1993; Kokubo & Ida 1998). As the low-mass planetesimals absorb the excess “heat” (or equivalent “angular momentum deficit”), they are captured by Saturn's (ν_6) secular resonance. Through resonant interaction, low-mass objects on the ν_6 resonance lose angular momentum while preserving their energy such that their eccentricity is excited while their semimajor axis is preserved (Chambers 2001). Eventually, the heated residual low-mass objects are cleared as they either crash into the Sun or are scattered and ejected by Jupiter when they cross its orbit. Since the ν_6 resonance is unique to the solar system configuration, these clearing and eccentricity damping mechanisms do not generally apply to planetary systems around other stars.

Another difficulty with the above scenario is that the timescale (~ 0.1 Myr) for the embryos interior to the snow line to attain their isolation masses is much shorter than the disk depletion timescale. The random motions of planetesimals and embryos are effectively damped by their tidal interaction with the disk gas (Artymowicz 1993; Ward 1993). Consequently, the embryos are likely to be well separated with severely depleted feeding zones (Kokubo & Ida 2002). Growth beyond the isolation-mass barrier requires the embryos' eccentricity excitation, which is possible when the surface density of the disk gas is depleted well below that of the minimum-mass nebula (Kominami & Ida 2002). However, subsequent damping is required to reduce the orbital eccentricity of the protoplanets to their present-day values. Protracted presence of gas can damp the prototerrestrial planets' eccentricity, but it also suppresses and delays their growth through coagulation.

The resolution of this dichotomy requires two independent processes for the protoplanetary embryos' eccentricity excitation and damping. Here we explore an avenue that may lead to a dynamical transition through a “shake-up” of planetary systems after the emergence of the “isolated” embryos, the formation of the gas giants, and a subsequent transition from chaos to order through protoplanet-disk tidal interaction (Lin et al. 2005). In the context of this scenario, we examine the following processes:

1. The gas giants exchange angular momentum with residual planetesimals and embryos through secular interaction. Since their energy is preserved, the embryos' eccentricities are modulated.
2. The effectiveness of secular interaction between gas giants and the residual embryos is suppressed by the gravitational potential of their nascent solar nebula until it is depleted below the surface density of the minimum-mass nebula.
3. The precession rates of the gas giants' orbits due to the disk potential decline with gas depletion. When these rates coincide with those induced by the gas giants on the embryos, their relative longitude of periastron becomes quasi-static and the embryos' eccentricities are greatly excited. The gas giants' secular resonances sweep across the inner region of the solar system as the residual gas in the disk vanishes.
4. The embryos' eccentricity is also damped by their tidal interaction with the residual gas near their orbits. Under the combined influence of the sweeping secular resonance and their interaction with the disk, embryos migrate inward.
5. The near-apsidal alignment of the orbit-crossing embryos enhances their close-encounter frequency and decreases their relative velocities. The low-impact collisions lead to the final assemblage of prototerrestrial planets.

6. Collisions/close encounters with residual embryos also dislodge prototerrestrial planets from the gas giants' secular resonances. Outside the resonances, the protoplanets' eccentricity is damped by their tidal interaction with the disk to the present-day kinematic configuration of the inner solar system.

We are primarily interested in whether these processes may induce planets to form with circular orbits and wide spacings. The damping of the planets' eccentricity would greatly lengthen the growth timescale for dynamical instability and orbit crossing in these systems. The sequence of processes 1–3 is called a sweeping secular resonance and was originally proposed by Ward et al. (1976). It was confirmed analytically (Heppenheimer 1980; Ward 1981) and numerically (Lecar & Franklin 1997; Nagasawa et al. 2000, 2001) that the secular resonances pass through the terrestrial planet region during the disk depletion that proceeds exponentially (see eq. [12]) on a timescale $\tau_n \sim 3\text{--}10$ Myr (Haisch et al. 2001).

In this paper and a companion paper (E. W. Thommes et al. 2005, in preparation), we focus on the analysis of these processes and demonstrate the feasibility of this dynamical shake-up scenario (DSS). Here we primarily discuss the effect of the sweeping secular resonance. The secondary effects of induced embryo collisions will be discussed in the companion paper. In order to provide a quantitative description of our model, we outline, in § 2, the relevant processes that determine the growth of planetesimals, protoplanetary embryos, and gas giant planets. Using analytic approximations, we show in § 3 how the secular resonance sweeps through the terrestrial planet region. The nonlinear response of the embryos is computed numerically. In § 4 we describe the numerical methods we used in our calculation. Through a series of numerical simulations, we examine the condition for embryo collision in §§ 5 and 6. Based on a linear approximation, we deduce analytically the condition under which embryos undergo resonant orbital migration in § 7. We also show, in §§ 5 and 7, that due to the combined influence of secular resonance and tidal drag, the longitudes of periastris of the migrating embryos are approximately aligned. This alignment promotes frequent collisions. Finally, in § 8 we summarize our results and discuss their implications.

2. THE SEQUENTIAL ACCRETION MODELS OF SOLAR SYSTEM FORMATION

2.1. Growth Rates and Isolation Mass

Planetesimals evolve through collisions, the frequency of which is determined by their mass (m), radius (R), and velocity dispersion (σ). In the limit that their $\sigma^2 < Gm/R$, the growth rate of the planetesimals is (Safronov 1969)

$$\begin{aligned} \frac{dm}{dt} &\sim \pi R^2 \rho_d \left(\frac{2Gm}{R\sigma^2} \right) \sigma \sim 2\pi R^2 \Sigma_d \Omega_K \left(\frac{Gm}{R\sigma^2} \right) \\ &\sim 2\pi \left(\frac{R}{a} \right) \left(\frac{m}{M_*} \right) \left(\frac{a\Omega_K}{\sigma} \right)^2 \Sigma_d a^2 \Omega_K, \end{aligned} \quad (1)$$

where $\Omega_K = (GM_*/a^3)^{1/2}$ is the Kepler frequency, G is the gravitational constant, M_* is mass of the central star, and $\rho_d (= \Sigma_d \Omega_K / \sigma)$ is the spatial density of particles. In a minimum-mass nebula, the surface density of heavy elements at radius a is

$$\Sigma_d \sim (f_d \eta_{\text{ice}}) 10 \left(\frac{a}{1 \text{ AU}} \right)^{-3/2} \text{ g cm}^{-2}, \quad (2)$$

where f_d is a scaling factor and $\eta_{\text{ice}} = 1$ and 4.2 on either sides of the snow line (Hayashi et al. 1985). [In the original version of

TABLE 1

NOTATIONS

Symbol	Meaning
g	Precession rate $\sim d\varpi/dt$
$g_{p,\text{disk}}$	Precession rate due to the disk (eq. [13])
$g_{p,J}$	$GM_J b_{3/2}^{(1)} / (4naa_J^2) \equiv t_{p,J}^{-1}$
M_{iso}	Isolation mass (eq. [4])
M_J	Jovian mass
m_e	Embryo's mass
μ_J	$ C_{p,J} e_J / t_{p,J} = GM_J b_{3/2}^{(2)} e_J / (4naa_J^2)$
n	Mean motion
ϖ	Longitude of pericenter
Σ_d	Dust surface density (eq. [2])
Σ_g	Gas surface density
τ_{damp}	Tidal damping timescale (eq. [8])
τ_n	Disk depletion timescale
u	$\exp(-t/\tau_n)$

the MMNM, $\Sigma_d(1 \text{ AU}) = 7 \text{ g cm}^{-2}$, which is equivalent to $f_d = 0.7$ in the above expression.] Important variables are tabulated in Table 1. By taking into account the effect of gas drag on the planetesimals' velocity dispersion σ , Kokubo & Ida (2002) derived the planetesimals' growth timescale to be

$$\begin{aligned} \tau_{\text{acc}} &\simeq 0.12 \eta_{\text{ice}}^{-1} f_d^{-1} f_g^{-2/5} \left(\frac{a}{1 \text{ AU}} \right)^{27/10} \\ &\times \left(\frac{m}{M_{\oplus}} \right)^{1/3} \left(\frac{M_*}{M_{\odot}} \right)^{-1/6} \text{ Myr}, \end{aligned} \quad (3)$$

where f_g is a scaling factor for the gas surface density with respect to that of the MMNM. Note that τ_{acc} is an increasing function of m because more massive planetesimals become embryos that stir up σ of their neighbors and reduce their own growth rate. This m dependence leads to a self-regulated oligarchic growth phase during which a population of comparable-mass embryos emerge in the midst of a swarm of small and relatively eccentric planetesimals (Kokubo & Ida 1998; Thommes et al. 2003). In this system, the orbital spacings of the embryos (equivalently, full width of their feeding zone, Δa) are nearly equal to one another and are as large as $\sim 10 r_H$, where $r_H = (m/3M_*)^{1/3} a$ is the Hill radius.

When they consume most of the planetesimals in their feeding zones, the embryos' growth is stalled (Lissauer 1987) with masses (m_e) comparable to the local isolation mass

$$\begin{aligned} M_{\text{iso}} &= 2\pi a \Delta a \Sigma_d \\ &\simeq 0.16 \eta_{\text{ice}}^{3/2} f_d^{3/2} \left(\frac{\Delta a}{10 r_H} \right)^{3/2} \left(\frac{a}{1 \text{ AU}} \right)^{3/4} \left(\frac{M_*}{M_{\odot}} \right)^{-1/2} M_{\oplus}. \end{aligned} \quad (4)$$

Growth beyond isolation requires the excitation of the embryos' eccentricity, which enables their orbits to cross. This process is possible after the gas depletion, which essentially eliminates its damping effects (Kominami & Ida 2002). In a gas-free environment, the velocity dispersion of the embryos grows until they cross each other's orbit (Chambers et al. 1996) on a timescale

$$\log \tau_x \simeq \frac{3\delta}{4} + \log \delta - \frac{1}{2} + \frac{1}{3} \log \left(\frac{m}{M_*} \right), \quad (5)$$

which is a rapidly increasing function of $\delta = 2^{-1/3} \Delta a / r_H$. The final stages of the embryos' growth may proceed through giant

impacts (Lissauer & Stewart 1993). Eventually a few surviving embryos acquire most of the residual planetesimals and less massive embryos during the late oligarchic growth stage. Since their collisional cross section is reduced to their geometrical size, the mass of the embryos (subscript e) evolves with a growth rate given by $dm_e/dt = m_e/\tau_{e,\text{acc}}$:

$$\tau_{e,\text{acc}} \simeq 20\eta_{\text{ice}}^{-1}f_d^{-1}\left(\frac{a}{1\text{ AU}}\right)^3\left(\frac{m_e}{M_{\oplus}}\right)^{1/3} \times \left(\frac{\rho_e}{1\text{ g cm}^{-3}}\right)^{2/3}\left(\frac{M_*}{M_{\odot}}\right)^{-1/2}\text{ Myr}. \quad (6)$$

From equations (3) and (6), we note that $\tau_{\text{acc}}(M_{\text{iso}}) < \tau_n < \tau_{e,\text{acc}}(M_{\text{iso}})$ interior to the snow line. Prior to gas depletion, the feeding zones are severely depleted; i.e., most planetesimals within them are retained by the embryos. Since $\tau_x(\delta = 10) + \tau_{e,\text{acc}} \geq \tau_n$, the embryos remain isolated after the gas depletion. But since gas giants must form in a gas-rich environment, isolation for terrestrial planets is overcome well after the emergence of the gas giants. Although the long growth timescales for the Earth (60 Myr) and for Mars (75 Myr) in equation (6) are comparable to the results of some numerical simulations (Chambers & Wetherill 1998), they are inconsistent with those deduced through the cosmochemical consideration (10–30 Myr for the Earth and 4–10 Myr for Mars; Yin et al. 2002). Also note that $\tau_x + \tau_{e,\text{acc}}$ increases with a , i.e., the final formation of a proto-planet proceeds from the inside to the outside of the disk. This inference is again inconsistent with the estimated age differences between Mars and the Earth (Jacobsen 2005).

2.2. Disk-Core Interaction

Goldreich & Tremaine (1980) showed that tidal interaction leads to an exchange of angular momentum between the embryos and their nascent disks. In a follow-up analysis, Ward (1986, 1997) showed that an isolated embedded embryo loses angular momentum to the disk exterior to its orbit at a rate faster than it gains from the disk interior to its orbit and this torque imbalance,

$$\Delta T \simeq \frac{c_1}{2}\left(\frac{m_e}{M_*}\right)^2\Sigma_g a^2(a\Omega_K)^2\left(\frac{a\Omega_K}{c_s}\right)^3, \quad (7)$$

may lead to an inward migration, which has come to be called “type I.” In the above equation Σ_g and c_s are the surface density and sound speed of the disk gas, respectively, and $c_1 \sim -0.25$ is the fractional imbalance in the torque. For an Earth-mass planetary embryo embedded at 5 AU in the minimum-mass nebula, the originally estimated migration timescale is 10^4 yr. Unless type I migration is suppressed, no gas giant planet can form through the core accretion process. Terrestrial planets in the solar system are overwhelmingly composed of refractory material, which indicates that their embryos formed inside the snow line and did not migrate significantly.

In a subsequent three-dimensional analysis, the derived magnitude of the torque imbalance is reduced from the expression in equation (7) by a factor of 2–3 (Tanaka et al. 2002). These estimates are based on the assumption of steady disk flow. But, in laminar disks, nonlinear dissipation near the embryos (Tanigawa & Watanabe 2002) can lead to local potential vorticity generation (Koller et al. 2003) followed by secondary instability (Balmforth & Korycansky 2001). The subsequent growth of vortices leads to large fluctuations in the effective torque applied by the embryos on the disk gas. If the embryos are embedded in a fully turbulent

disk, the unsteady flow of the disk gas near these embedded embryos further retards their radial migration (Rice & Armitage 2003; Nelson & Papaloizou 2004; Laughlin et al. 2004). Finally, secular interaction between the disk and several adjacent embryos may lead to shock dissipation that disrupts the wave patterns they excite in the gaseous disk (Bryden et al. 2000). In general, a large torque reduction suppresses the extent of type I radial migration.

Embedded embryos’ eccentricities are also damped and excited through their individual interaction with the disk at their Lindblad and corotation resonances, respectively (Goldreich & Tremaine 1980). Whereas type I migration is regulated by the balance offset between the embryos’ tidal torque on the inner and outer parts of the disk, both parts contribute additively to the eccentricity damping process on a timescale (Ward 1993; Artymowicz 1993)

$$\tau_{\text{damp}} \simeq \left(\frac{m_e}{M_{\odot}}\right)^{-1}\left(\frac{\Sigma_g a^2}{M_{\odot}}\right)^{-1}\left(\frac{c_s}{a\Omega_K}\right)^4\Omega_K^{-1}. \quad (8)$$

The relatively small τ_{damp} generally induces embryos to retain nearly circular orbits. Note that as the disk is depleted, τ_{damp} increases. When the disk gas is mostly depleted (with $f_g < 0.1$), it is possible for $\tau_{\text{damp}} > \tau_x$ so that the embryos’ eccentricity grows. Also note that the damping timescale is shorter for the more massive embryos such that the eccentricity of the merger products may be mostly damped. Through a series of numerical simulations, Kominami & Ida (2002, 2004) showed that there is a very limited window of opportunity for the low-mass embryos to attain sufficient velocity dispersion, to overcome isolation, and to merge (with $\sigma > \Delta a/2$) before the disk is completely depleted. The damping of the high-mass embryos’ eccentricities to their present values ($e < 0.05$) requires a protracted preservation of gas, whereas their emergence needs a relatively fast depletion of the disk. It remains unclear whether the relatively stable nearly circular orbits of the terrestrial planets can be established through a delicate timing of disk evolution.

2.3. The Dynamical Shake-up Scenario

In an attempt to resolve this paradox, we suggest that the embryos’ eccentricity was excited and damped by *two independent* processes. In this model, we assume that (1) in the presence of the gas, the embryos become isolated with small eccentricities; (2) after the emergence of the gas giants, the nearby embryos are strongly perturbed by the giants’ gravity; (3) during the epoch of gas depletion, the eccentricities of the distant embryos are also excited through the gas giants’ sweeping secular resonances; (4) the combined effects of the gas giants’ secular perturbation and embryo-disk tidal interaction lead to quasi-stationary aligned longitudes of periaapse among the resonant embryos; (5) the embryos’ overlapping and coherently oriented orbits promote their coagulation and further growth; (6) the merger products are dislodged from the gas giants’ secular resonances; and finally (7) their eccentricities are damped by their tidal interaction with the residual disk gas.

In this scenario, the σ growth of the dynamically isolated embryos is induced externally by the perturbation of the gas giants. We separate the eccentricity damping process. Both of these processes are expected to occur naturally during the final evolution of the disk gas. In response to two independent processes, a quasi-equilibrium may be attainable. The asymptotic results depend on the magnitude of (1) the gas giants’ eccentricities, (2) τ_n , and (3) f_d (through M_{iso}).

3. SWEEPING SECULAR RESONANCES

For a given semimajor axis, a planet attains a maximum in angular momentum when its orbit is circular. In an eccentric orbit, a planet possesses an “angular momentum deficit.” In multiple-planet systems, the angular momentum deficit of one planet can modulate the eccentricity of others through their secular interaction. The magnitude of the planets’ mutual perturbation and angular momentum exchange rate increases with the magnitude of their eccentricity. But the modulation amplitude of nonresonant planets is limited until the precession induced by the disk potential is suppressed by its depletion.

3.1. Physical Basis of the Secular Resonance

To the lowest order in eccentricity, the perturbation of a gas giant planet j (with semimajor axis a_j , mean motion n_j , eccentricity e_j , and longitude of periastron ϖ_j) introduces a precession of the periastron (Murray & Dermott 1999) of an embryo i (with semimajor axis a_i , mean motion n_i , eccentricity e_i , and longitude of periastron ϖ_i) at a rate

$$\frac{d\varpi_{i,\text{planet}}}{dt} = \sum_{j \neq i} \frac{1}{t_{i,j}} \left[1 + C_{i,j} \left(\frac{e_j}{e_i} \right) \cos(\varpi_i - \varpi_j) \right], \quad (9)$$

where

$$t_{i,j}^{-1} \equiv \frac{Gm_j \alpha b_{3/2}^{(1)}}{4n_i a_i^2 \max(a_i, a_j)} \equiv g_{i,\text{planet}} \quad (10)$$

is a secular part of the precession of planet i due to planet j , $C_{i,j} \equiv -b_{3/2}^{(2)}/b_{3/2}^{(1)}$, and $b_j^{(k)}$ are the Laplace coefficients whose variables are $\alpha = \min(a_i, a_j)/\max(a_i, a_j)$. In the context of solar system formation, while Jupiter and Saturn undergo secular interaction with each other, the embryos’ masses are generally too small to induce any significant precession in either of these gas giants.

Under the influence of the gravitational potential of the disk, the embryos and gas giants also undergo precession. In this paper we consider an axisymmetric thin disk whose initial surface density of the gas is

$$\Sigma_{\text{init}} = \Sigma_0 \left(\frac{a}{1 \text{ AU}} \right)^{-k}. \quad (11)$$

In the minimum-mass disk model, $\Sigma_0 = 1700 \text{ g cm}^{-2}$ and $k = 3/2$. In order to mimic the depletion of the disk, we consider the exponential decay of the gas surface density such that

$$\Sigma_g = \Sigma_{\text{init}} \exp\left(-\frac{t}{\tau_n}\right), \quad (12)$$

where τ_n is a disk depletion timescale. An initial apsidal precession rate of gas giant planet j due to this contribution is (Ward 1981)

$$g_{j,\text{disk}} \equiv \frac{d\varpi_{j,\text{disk}}}{dt} \sim -Z_k \left[\frac{\pi G \Sigma_{\text{init}}(a_j)}{n_j a_j} \right], \quad (13)$$

where Z_k is a constant of order unity for $-1 < k < 4$. In the case of a minimum-mass disk, $Z_{3/2} = 1.094$. During the depletion of the disk,

$$g'_{j,\text{disk}} = g_{j,\text{disk}} \exp\left(-\frac{t}{\tau_n}\right). \quad (14)$$

The disk potential also induces the embryos to precess at similar rates that are determined by the surface density in the vicinity of their orbits.

In the minimum-mass nebula, both $g_{j,\text{disk}}$ of the gas giants and $g_{i,\text{disk}}$ of the embryos are $>g_{i,\text{planet}}$ of the embryos induced by Jupiter and Saturn so that total precession rates of both populations are dominated by the disk potential. But during the depletion of the disk, the magnitudes of these precession rates pass through an epoch when they are comparable to each other. In addition, it is possible for the disk to induce Jupiter and Saturn to precess at rates similar to that imposed by the gas giants on the embryos. Embryos with $g'_{i,\text{disk}} = g_{i,\text{planet}}$ (due to the gas giants) experience a secular resonance in which the phase $(\varpi_i - \varpi_j)$ can be locked onto some nonvanishing values over a timescale much longer than $t_{i,j}$.

The magnitude of Σ_g needed for the embryos to experience the secular resonance depends on their location, but it is independent of their masses. As Σ_g declines, the location of the secular resonance propagates toward the Sun. This process is commonly referred to as sweeping secular resonance and has been attributed as a mechanism to excite the eccentricity of the asteroids (Ward et al. 1976; Heppenheimer 1980; Ward 1981; Lecar & Franklin 1997; Nagasawa et al. 2000)

3.2. Conditions of the Secular Resonance

Within the limit of the lowest order in eccentricity, the tidal torque induced by the planets on the disk does not modify the rate of their perihelion precession. The embryos can be considered massless bodies compared with the disk and Jovian planets. Therefore, the critical condition for the secular resonance is expressed with the same equations described by Heppenheimer (1980) and Nagasawa et al. (2000).

The secular resonances induced by Jupiter and Saturn occur when the apsidal precession rate of the embryos and the eigenfrequencies of the system coincide with each other. Interior to the Jovian orbit, the apsidal precession rate of an embryo p is

$$g_p = g_{p,J} + g_{p,S} + g_{p,\text{disk}} \exp\left(-\frac{t}{\tau_n}\right). \quad (15)$$

The eigenfrequencies of the system are dominated by the precession of Jupiter and Saturn such that

$$g_{5,6} = \frac{1}{2} \left[g_J + g_S \pm \sqrt{(g_J - g_S)^2 + 4g_{J,S}^2} \right], \quad (16)$$

where

$$g_J = g_{J,S} + g_{J,\text{disk}} \exp\left(-\frac{t}{\tau_n}\right), \quad (17)$$

$$g_S = g_{S,J} + g_{S,\text{disk}} \exp\left(-\frac{t}{\tau_n}\right), \quad (18)$$

$$g_{J,S} = -\frac{Gb_{3/2}^{(2)}(a_J/a_S)}{4a_S^3} \sqrt{\frac{m_J m_S}{n_J n_S}}, \quad (19)$$

where $g_{J,S} = t_{J,S}^{-1}$ and $g_{S,J} = t_{S,J}^{-1}$ are the precession of Jupiter/Saturn due to Saturn/Jupiter, respectively. Embryos encounter a secular resonance with Jupiter ν_5 when its $g_p = g_5 \sim g_J$. From equations (15) and (17), the resonant condition becomes

$$g_{p,J} + g_{p,S} - g_{J,S} = (g_{J,\text{disk}} - g_{p,\text{disk}}) \exp\left(-\frac{t}{\tau_n}\right). \quad (20)$$

Using $k = 3/2$ and the current mass and orbital values of Jupiter and Saturn, we estimate the time when ν_5 passes through $1 \text{ AU} < a < 3 \text{ AU}$ to be

$$u_R \equiv \exp\left(-\frac{t_R}{\tau_n}\right) = \frac{g_{p,J} + g_{p,S} - g_{J,S}}{g_{J,\text{disk}} - g_{p,\text{disk}}} \sim 6 \times 10^{-3} \left(\frac{a}{1 \text{ AU}}\right)^4. \quad (21)$$

Outside 3 AU, Jupiter's mean motion resonances induce modest perturbation on the embryos. When the ν_5 resonance propagates interior to 1 AU, the disk mass is so low that the precession of Jupiter is dominated by the perturbation of Saturn rather than the gravity of the disk. The corresponding timescale for the propagation of the resonant semimajor axis a_{res} is

$$\tau_{\text{res}} \equiv \left| \frac{a_{\text{res}}}{\dot{a}_{\text{res}}} \right| \simeq 4\tau_n. \quad (22)$$

3.3. Dependence on the Powers of the Surface Density

For a power-law Σ_g distribution, the time of the secular resonance passage does not sensitively depend on the power index if it is within the range $1 < k < 5/2$. Figure 1 shows the relation of $\exp(-t/\tau_n)$ and the location of the secular resonance for different values of k . Here we use the $\Sigma_0 = 1700 \text{ g cm}^{-2}$ and derived the value of Z_k to be $Z_1 = 1$, $Z_{3/2} = 1.094$, $Z_2 = 1$, and $Z_{5/2} = 0.685$.

If the effect of damping due to embryo-disk interaction is neglected, the intensity of the secular resonance is inversely proportional to the square root (Heppenheimer 1980) of the changing rate of the resonant frequency (i.e., $\propto [d(g - g_l)/dt]^{-1/2}$, where $l = 5$ and 6 for the ν_5 and ν_6 resonance, respectively; Ward 1981; Nagasawa et al. 2000). Since the time evolution of the ν_5 location is insensitive to the values of k , there is also little difference in the strength of the resonance when it propagates through the terrestrial planet region. But for a very flat Σ_g distribution, the difference in the disk-induced precession frequencies between the gas giants ($g_{J,\text{disk}}$ and $g_{S,\text{disk}}$) and the embryos ($g_{i,\text{disk}}$) is reduced such that the secular resonances no longer sweep across the terrestrial region when the surface density decreases below 1700 g cm^{-2} . Although we do not know in general the detailed local structure of the disk near the protoplanets, the discussions based on the results of $k = 3/2$ would be appropriate as long as the power of the surface density is in a range of $1 < k < 5/2$. This range contains not only the MMNM but also an evolving disk model with an α -prescription for turbulence (Lin & Bodenheimer 1982).

3.4. Dependence on the Disk Structure near the Gap

Equation (4) indicates that outside the snow line the embryos' isolation mass M_{iso} becomes comparable to a few Earth masses and embryos evolve into protoplanetary cores as the disk gas accumulates into static envelopes around them. Initially, low-mass cores are surrounded by thin gaseous envelopes that are heated at their base by the release of gravitational energy from the contracting gas and the impinging residual planetesimals. The replenishment of the envelope leads to the accretion of additional gas at a rate that is determined by the heat through its outer radiative layer (Pollack et al. 1996). If dust grains can coagulate and sediment so that the grain opacity is reduced from its values in the interstellar medium (Podolak 2003), the envelopes will be able to attain masses comparable to their cores within 3 Myr, i.e., prior to severe disk depletion. Thereafter, gas in the envelope contracts

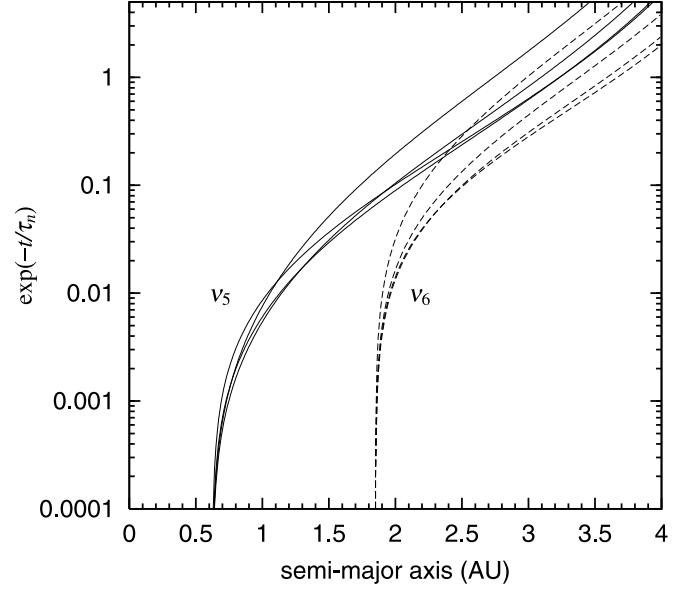


FIG. 1.—Location of the ν_5 and ν_6 resonances depending on the different power ($k = 1, 3/2, 2$, and $5/2$) of the surface density of the gas disk $\Sigma_{\text{init}} \propto 1700(a/1 \text{ AU})^{-k} \text{ g cm}^{-2}$.

rapidly onto the cores and together they evolve into gas giant planets (Stevenson 1982b; Mizuno 1980).

A protoplanet's mass increases until it is sufficiently large to tidally perturb the disk structure by exciting density waves that carry negative/positive angular momentum at their inner/outer Lindblad resonances (Goldreich & Tremaine 1980). This deficit/excess angular momentum is deposited into the gas as the waves dissipate during their propagation inward/outward from the resonances. If the rate of tidal angular momentum transport exceeds that due to turbulent viscosity and the tidal radius of the planet exceeds the disk thickness, a gap will form in the vicinity of the protoplanet's orbit (Lin & Papaloizou 1993; Takeuchi et al. 1996).

Formation of this gap is important to the direction of apsidal precession induced by the disk potential on the Jovian planets and embryos. The embryos are fully embedded in the disk without any gaps around them. The mean contribution to the embryos' apsidal precession $g_{j,\text{disk}}$ is dominated by the disk regions near their orbits. In this limit, we find from equation (13) $g_{j,\text{disk}} < 0$.

Around a gas giant with a sufficiently large mass, a gap is formed with a radial extent between a_1 and a_2 that is larger than the maximum of $2r_H$ and $2ea$. In contrast to the embryos, the disk potential in the immediate vicinity of the gas giant's orbit is excluded. The initial apsidal precession $g_{j,\text{disk}}$ is positive with a magnitude

$$g_{j,\text{disk}} = \frac{d\varpi}{dt} \sim \left(\frac{\pi G \Sigma_0}{n_j a_j} \right) \sum_{l=1}^{\infty} l(2l+1) A_l \times \left[\frac{(a_1/a)^{2l+2-k}}{2l+2-k} + \frac{(a/a_2)^{2l-1+k}}{2l-1+k} \right], \quad (23)$$

where $A_l = [(2n)!/2^{2n}/(n!)^2]^2$. During the disk depletion, the apsidal precession rate decreases and its magnitude becomes $g'_{j,\text{disk}} = g_{j,\text{disk}} \exp(-t/\tau_n)$.

Despite this directional difference in the precession induced by the disk potential, secular resonance between the gas giants (in eq. [23]) and the embryos (in eq. [13]) occurs because the

embryos' total precession frequency g_p is also influenced by the secular perturbation of the gas giants (see eq. [15]). After the disk mass is reduced below that of gas giants, g_p of librating embryos becomes episodically positive as it becomes dominated by the secular perturbation of the gas giants through $g_{p,J}$ and $g_{p,S}$ (see eq. [9]). (Since $|C_{i,j}| > 1$ for small e_i , embryos on low-eccentricity orbits have a tendency to librate.) The secular interaction between Jupiter and Saturn also modulates their $g_{J,S}$ as their relative longitude of periastron $\varpi_J - \varpi_S$ varies. Taking these effects into account, we show both analytically (see below) and numerically (in Fig. 2) the time and the location of the Jovian secular resonance in the case that the minimum-mass disk has a gap between $a_J(1 - \Delta d)$ and $a_J(1 + \Delta d)$ and $a_S(1 - \Delta d)$ and $a_S(1 + \Delta d)$. In this figure we examine the cases of $\Delta d = 0, 0.2, 0.3, 0.4$, and 0.5 . In all cases, the sweeping secular resonances occur with nearly the same values of Σ_g and on very similar timescales despite the gap formation in the disk.

In the absence of a gap, the secular resonance sweeps the region of < 2 AU when the disk mass reduces below 10% of its initial minimum-mass nebula value. The formation of the gap does not significantly modify the time of the sweeping secular resonance's passage except for the regions near the gap. The timing of the resonance passage is delayed by $\sim 0.3\tau_n$ at 1 AU and $\sim 0.8\tau_n$ at 2 AU. The strength of the resonance hardly changes, since it depends on the speed of the resonance passage and Jovian eccentricity rather than the disk mass.

3.5. Absence of Saturn

The sweeping secular resonance also occurs with only one Jovian planet. The main difference in this case is that the planet's precession is determined solely by the disk potential. **As we indicated in equation (23), $g_J > 0$ after it opens a gap near its orbit.** Despite the opposite signs between the planet's $g_{J,disk}$ and the embryos' $g_{i,disk}$, the embryos are also perturbed by the planet. Through the contribution of $g_{p,J}$, the precession of low-eccentricity librating embryos g_p undergoes episodic sign changes as the disk is depleted.

In the solar system, the mass of Jupiter (M_J) is much larger than that of Saturn (M_S) and $a_J < a_S$. Jupiter's (ν_5) secular resonance sweeps inward ahead of that of Saturn (ν_6) with a much greater strength. Nevertheless, Saturn eventually induces sufficient perturbation to stall the inward propagation of ν_5 just interior to the present-day semimajor axis of Venus.

Around some stars (such as τ Boo), single massive planets are found with modest to large eccentricities. In these systems, the sweeping secular resonance can propagate to a few stellar radii where the effect of the embryos' precession is modified by the relativistic correction to their host stars' post-Newtonian gravity.

The strength of the sweeping secular resonance is determined by the planet's eccentricity. Around stars with single planets on a circular orbit, the secular resonance cannot significantly perturb embryos in the disk. But around stars with highly eccentric planets, sweeping secular resonance may greatly excite the eccentricity of embryos over an extended region. Combined with the influence of tidal gas drag (see below), this causes embryos to be cleared out of these regions.

4. NUMERICAL METHODS OF ORBITAL EVOLUTION CALCULATIONS

The effects of sweeping secular resonances on (asteroidal) bodies have already been demonstrated with analytic methods and numerical calculations elsewhere (Heppenheimer 1980; Ward 1981;

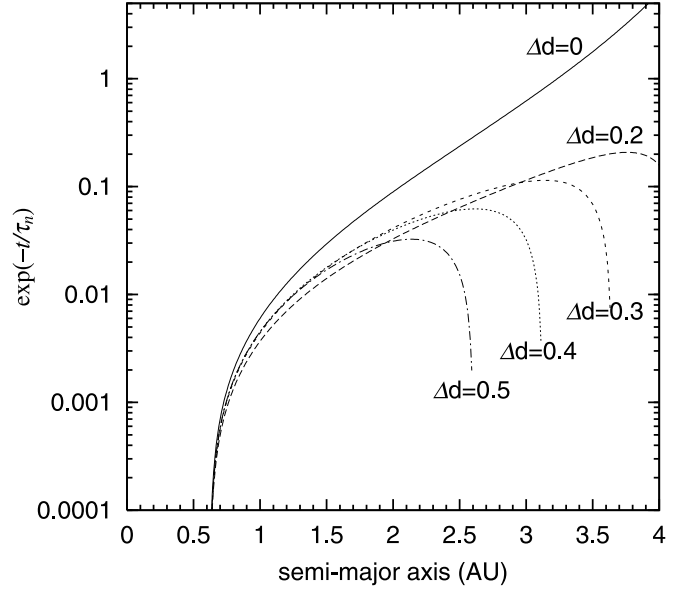


FIG. 2.—Location of the ν_5 resonance in the case where the Jovian planets have gaps. The edges of the gaps are located at $a_J(1 \pm \Delta d)$ and $a_S(1 \pm \Delta d)$.

Lecar & Franklin 1997; Nagasawa et al. 2000). Here we consider the additional effect of eccentricity damping due to the embryos' tidal interaction with the disk.

First, we carry out a set of numerical computations on the orbital evolution of embryos with isolation masses under the secular perturbation of both Jupiter and Saturn. We assume that the embryos and the gas giants form a coplanar system and the latter have eccentric orbits. The depleting disk both exerts eccentricity damping and contributes to the gravitational potential. We also assume that the effect of the disk exponentially decays on a timescale τ_n . Note that τ_n is an e -folding time, not the time by which the disk completely disappears.

In our numerical computation, we assume that the embryos do not open a gap, which is appropriate for the parameters we are considering. Embryos with isolation mass less than an Earth mass orbiting a solar-mass host star at a few AU radius cannot severely clear a gap in a disk that is heated by the stellar irradiation. For simplicity, we consider that the disk has spread outside the Saturnian orbit smoothly (i.e., no gaps around Jovian planets). As per the discussion in § 3.4, the timing of the sweeping secular resonance is not strongly affected by the disk structure and our approximation does not introduce any significant difference in our results.

The minimum-mass disk's contribution to the gravity at the location of the embryos can be determined by (Heppenheimer 1980; Ward 1981)

$$f_{i,disk} = -1.094(4\pi)G\Sigma_{init} \exp\left(-\frac{t}{\tau_n}\right), \quad (24)$$

where the surface density of the gas Σ_{init} in equation (11) corresponds to Σ_d in equation (2). In this paper we set the scaling parameter $f_g = 1$. We also assumed a thin axisymmetric disk structure, since the precession and eccentricity evolution of the embryos do not depend sensitively on the nonaxisymmetric torque induced by the planet-disk interaction. In modeling type I planet-disk interaction, we also neglect migration due directly to disk torques and apply only damping of eccentricities (see discussions in § 2.2).

For the numerical computation of the effect of embryo-disk tidal interaction (Ward 1993), we adopt the same approximation formula as Kominami & Ida (2002),

$$f_{i,\text{drag}} = -\frac{(v - v'_{\text{Kep}})}{\tau_{\text{damp}}} \exp\left(-\frac{t}{\tau_n}\right), \quad (25)$$

where v'_{Kep} is the velocity of the circular motion and τ_{damp} is given in equation (8). In our models we use

$$\tau_{\text{damp}} = 1.5 \times 10^{-3} \left(\frac{m}{M_{\odot}}\right)^{-1} \left(\frac{a}{1 \text{ AU}}\right)^2 \text{ yr}. \quad (26)$$

The rate of eccentricity change due to this drag is

$$\tau_e = \frac{1}{e} \frac{de}{dt} = -\frac{1}{\tau_{\text{damp}}} \exp\left(-\frac{t}{\tau_n}\right). \quad (27)$$

With this prescription, the eccentricity of unperturbed embryos evolves as

$$e = e_0 \exp \left\{ \frac{\tau_n}{\tau_{\text{damp}}} \left[\exp\left(-\frac{t}{\tau_n}\right) - \exp\left(-\frac{t_0}{\tau_n}\right) \right] \right\}. \quad (28)$$

The above equation indicates that e would become vanishingly small if the disk depletes on a timescale much longer than that for eccentricity damping. But if $\tau_n \ll \tau_{\text{damp}}$, the initial eccentricity of the embryos e_0 would be preserved.

We take into account the effect of the modification of v'_{Kep} from perfectly Keplerian rotation due to the disk potential such that, at a given radial location r ,

$$v'_{\text{Kep}} = \sqrt{\frac{|f_{i,\text{disk}}|}{r} + \frac{G(m + M_{\odot})}{r^3}}. \quad (29)$$

In the frame centered on the Sun, the gravitational force on the planet i , located at r_i , is

$$f_i = -\frac{G(M_{\odot} + m)}{r_i^3} r_i + G \sum_{j \neq i} m_j \left(\frac{r_j - r_i}{|r_j - r_i|^3} - \frac{r_j}{|r_j|^3} \right) + f_{i,\text{disk}} + f_{i,\text{drag}}. \quad (30)$$

With this prescription for the force, we compute the evolution of embryos' orbit using the full two-dimensional equation of motion. We used a Hermite integration and a time reversal Encke's method to compute the embryos' evolution. Typical models are computed for 100 Myr.

5. ORBITAL EVOLUTION OF AN EMBRYO DURING THE DISK DEPLETION

In order to take into account the secular perturbation due to both Jupiter and Saturn, as well as collisions between embryos, it is convenient to utilize the results of numerical simulations that are obtained with the methods described in § 4. We present here the results of various models to illustrate the overall trend of the embryos' orbital evolution in the presence of Jupiter, Saturn, and a depleting disk. We show here dynamical studies of the long-

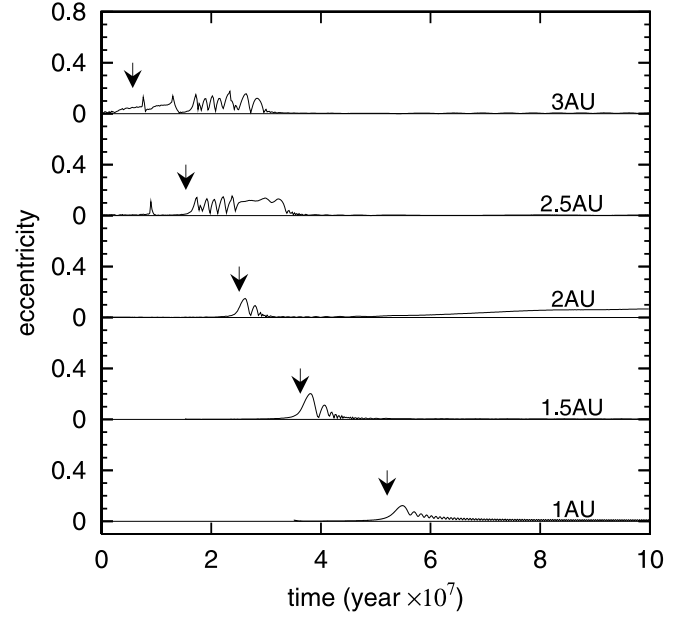


FIG. 3.—Timing of the secular resonance passage at different semimajor axes obtained from the integration of the full equations of motion. Arrows show the analytically estimated time of the passage (eq. [49]). Values of $e_j = 0.01$, $\tau_n = 10$ Myr, and $m = 0.03 M_{\oplus}$ are used.

term evolution of residual embryos as a consequence of sweeping secular resonance and embryo-disk tidal interaction by direct N -body simulations.

5.1. Passage of the Secular Resonance

The location of the ν_5 resonance is essentially determined by the mass and the semimajor axes of Jupiter and Saturn, as well as the mass of the disk. Figure 3 shows the timing of the resonance passages at different semimajor axis. Each line corresponds to the eccentricity evolution of one embryo ($0.03 M_{\oplus}$) perturbed by Jupiter and Saturn during the disk depletion ($\tau_n = 10$ Myr). For presentation purposes, we use the $e_j = 0.01$ in order to avoid the strong embryo's migration (see §§ 5.3 and 7.4). The arrows indicate the instance of the ν_5 secular resonance passage, which is derived analytically with the approach that is described in § 3.2. The timing of the resonance passage is independent of the value of Jovian eccentricity.

The results in Figure 3 indicate that the analytical solutions of § 3.2 reproduce the numerical computations well. The secular resonance sweeps from outside to inside. The embryos' eccentricity is excited by the secular resonance and is damped by the tidal drag. There are many second and higher order mean motion resonances outside 2 AU. The strength of these resonances depends on the eccentricity of both Jupiter and the embryos (Murray & Dermott 1999). The embryos initially at 2.5 and 3 AU are affected by both secular resonance and mean motion resonances, and their eccentricities vary after the passage of the secular resonance. The eccentricity of an embryo at 2 AU is slightly excited after the disk depletion because this location is close to the ν_6 .

The strength of the ν_5 resonance is determined by e_j . In this set of models, it is weak because we have adopted $e_j = 0.01$ for illustration purposes. Consequently, the embryos' eccentricity excitation is limited. The corresponding migration rate is also smaller than the propagation speed of the ν_5 secular resonance so that the embryos in the inner solar system are not caught in resonance (see § 7.4). Consequently, these embryos do not

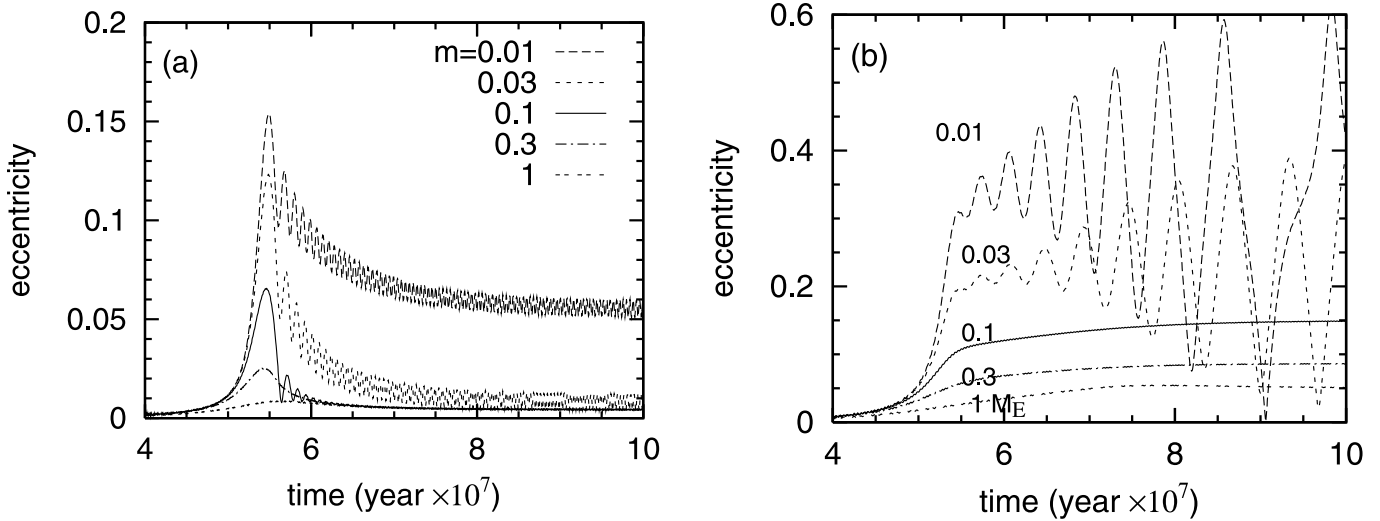


FIG. 4.—Evolution of an isolated embryo's eccentricity during the passage of a sweeping secular resonance. The initial semimajor axes are 1 AU in all cases. The long-dashed, short-dashed, solid, dot-dashed, and double-dashed lines show $m = 0.01, 0.03, 0.1, 0.3$, and $1 M_{\oplus}$, respectively. The Jovian eccentricity is 0.01 in panel (a) and 0.048 in panel (b). The disk depletion timescale is $\tau_n = 10$ Myr.

migrate inward over an extended region of the disk, and their eccentricity is damped after the passage of the resonance.

5.2. Eccentricity Excitation by Sweeping Secular Resonance and Damping by Tidal Drag

In order to consider the protracted trapping of embryos by the sweeping secular resonance, we compare two series of models with $e_J = 0.01$ and 0.045 , which is the present eccentricity of Jupiter. The depletion timescale is set to be 10 Myr. Typical variation of the eccentricity is shown in Figures 4a and 4b ($e_J = 0.01$ and 0.045 , respectively). For these models, we consider a collection of noninteracting embryos with various masses. In all cases, we place the embryos at the same semimajor axis, 1 AU.

The asymptotic eccentricity of embryos with mass less than $0.03 M_{\oplus}$ is relatively large (~ 0.05) and the eccentricities of larger embryos are damped after the passage of the resonance in the case of Figure 4a (see eq. [54]). In the case of Figure 4b, the embryos' eccentricity is maintained at relatively large values. The excitation magnitude of eccentricity is directly determined by the Jovian eccentricity (see § 7.4). Above an eccentricity threshold, the resonant capture condition in equation (59) can be satisfied. The protracted maintenance of secular interaction promotes the excitation of the embryo's eccentricity. The results in Figure 4b also indicate that sweeping secular resonance can lead to clearing of sub- $M_{\oplus}$ embryos at the 1 AU region provided that Jupiter has already acquired most of its present mass and eccentricity.

The secular resonance passes through the region of 1 AU at around 50 Myr ($5\tau_n$). Due to Jupiter's secular perturbation on Saturn, the ν_6 does not venture to 1 AU. During the passage of the sweeping secular resonance, the eccentricity of the embryos increases. The required eccentricity to cause orbit crossing of isolated protoplanets that are separated by $10r_H$ is 0.05 or less around 1 AU.

Since the mass of the embryos is small compared to Jupiter, the extent of secular excitation of eccentricity is independent of their mass. But our numerical results indicate that the excitation magnitude of the eccentricity is inversely proportional to the embryos' mass. The main cause for this mass dependence is through the rate of damping due to the embryos' tidal interaction with the disk.

5.3. Evolution of the Semimajor Axis

The embryos lose angular momentum but preserve their orbital energy during their secular interaction with Jupiter. In contrast,

tidal interaction between embedded embryos and the disk damps the embryos' epicyclic motion without a significant net change in their orbital angular momentum. These processes together reduce the semimajor axes of the embryos and cause them to migrate (Thommes et al. 2003). In this section we present numerical results to demonstrate that this self-regulated migration process can lead to a protracted resonant interaction between the gas giants and residual embryos.

The current location of the ν_5 is about 0.6 AU. In a minimum-mass nebula, it is about 3 AU (see Fig. 1). During the global depletion of the disk, the resonance location sweeps from the outer disk inward. We find, from equation (22), that the timescale associated with the propagation of the ν_5 resonance is comparable to the disk depletion timescale. From the observation of the T Tauri stars, $\tau_n \sim 3\text{--}10$ Myr. The damping timescale of the semimajor axis due to the tidal drag is much shorter than τ_n for the protoplanets. In the absence of mutual interaction with their siblings, the eccentricity of embryos with mass comparable to the isolation mass is effectively damped when they are not in the gas giants' secular or mean motion resonances. But when they are locked into a sufficiently strong sweeping secular resonance, embryos can attain adequate eccentricity to sustain an inward migration along with the inwardly sweeping secular resonance.

We illustrate how the sweeping secular resonance captures the protoplanets in Figure 5. The dashed lines represent the location of the perihelion and aphelion, i.e., the orbital excursions. The thick solid line shows the location of the ν_5 . The embryos were initially located at three different semimajor axes (3, 1.5, and 1 AU). In the outer (>1.5 AU) region, the forced eccentricity induced by Jupiter's secular perturbation leads to the embryos' migration prior to the arrival of the ν_5 secular resonance. As the ν_5 resonance sweeps inward, it captures embryos with modest initial eccentricity. The embryos' eccentricities increase gradually and then reach some equilibrium value as their semimajor axes decrease. The asymptotic value of the eccentricity is decreased with the embryos' mass (see eq. [67]).

In addition to the present value of Jupiter's eccentricity, we also considered a case with $e_J = 0.01$. The results are generally similar to the previous cases. But with a weaker secular resonance, the embryos' migration, starting at an initial radius of 3 AU, is stalled at Jupiter's 3:1 resonance (2.5 AU) for the $m_e = 0.3 M_{\oplus}$ case. Embryos that started at 1.5 and 1 AU initially are not

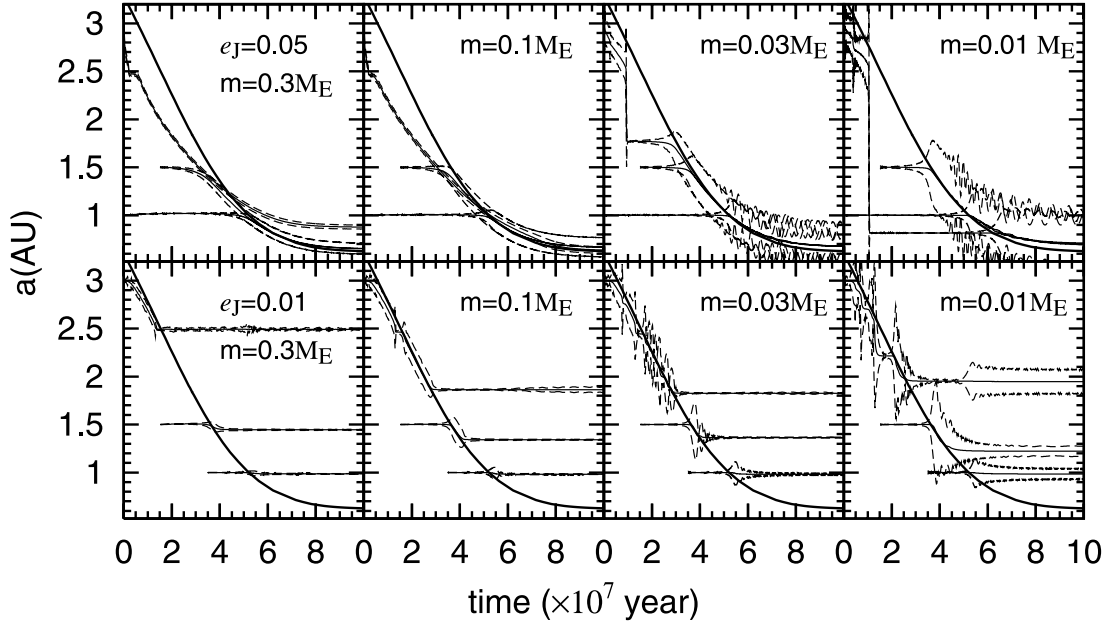


FIG. 5.—Evolution of semimajor axes (solid line) and perihelia and aphelia (dashed lines). The thick solid line shows the ν_5 resonance. Top and bottom panels are for $e_J = 0.05$ and 0.01 , respectively. Masses of the embryos are 0.3 , 0.1 , 0.03 , and $0.01 M_\oplus$ from left to right.

caught by the sweeping secular resonance. For the low-mass cases, the less efficient eccentricity damping promotes more rapid migration such that the migration of those with initial $a = 3$ AU is stalled interior to 2 AU. Low-mass embryos with relatively small (1.5 and 1 AU) initial a undergo very limited amount of migration before they become detached from the secular resonance. The insensitivity of the stalling radii to m_e is consistent with the analytic solution (see eq. [73]).

5.4. Locking of the Relative Angle of Perihelion

In § 7.4.2 we show that it is possible to capture embryos onto a gas giants' secular resonance when their $\Delta g < \Delta g_{\text{crit}}$ (see eq. [62]). Since this condition is easily attained with small eccentricities, embryos with initial circular orbits can be affected by the sweeping secular resonance provided that the gas giants' e_J is sufficiently large.

The results in Figure 6 indicate that in all cases, the embryos first enter into the secular resonance with $\eta = \pi$ (see discussion in § 7.4.2), where η is the relative longitude of embryos and Jupiter. As the disk depletion reduces the magnitude of Δg just below that of Δg_{crit} (eq. [62]), $d\eta/dt$ vanishes as η increases. During the subsequent evolution, the embryos' η -values evolve until a quasi-equilibrium is achieved (i.e., the right-hand side of eq. [58] vanishes). Combining either equation (67) or equation (71) with equation (63), we find $\delta\eta \propto m_e^{2/3}$ (see eq. [75]) when the embryos are trapped in the secular resonance. For $m_e < 0.3 M_\oplus$, the magnitude of $\delta\eta$ is generally much smaller than unity during their trapping such that the longitudes of their periape are essentially aligned. This analytic result is in good agreement with our numerical computation for which we show in Figure 6 the evolution of embryos with various masses. The initial conditions for these calculations are identical to those in Figure 4b.

5.5. Disk Depletion Timescale

We now consider the dependence of e and the radial extent Δa of migration on τ_n . In Figure 7 we consider four values of τ_n (0.5 , 1 , 5 , and 10 Myr) and m_e (0.03 , 0.1 , 0.3 , and $1 M_\oplus$). In all cases, the initial value of a is set to be 1 AU. For the long

depletion timescale ($\tau_n = 5$ and 10 Myr) cases, the embryos are trapped by the sweeping resonance and they migrate by a radial extent $\Delta a_{\text{mig}} \sim 0.3$ AU and are stalled near the present location of the ν_5 resonance. During this migration, the embryos' eccentricity attains equilibrium values that agree well with those in equations (71) and (67), i.e., $e \propto m_e^{-1/3}$.

But for the short depletion timescale ($\tau_n = 0.5$ and 1 Myr) cases, Δa_{mig} values of the embryos are negligibly small and are not caught by the sweeping secular resonances. After an initial excitation during the passage of the resonance, the embryos' eccentricity reaches values that are decreasing functions of m_e (see eq. [52]). Thereafter, e declines on a timescale $\tau_{\text{damp}} \sim m_e^{-1}$ (see eq. [26]).

These numerical results are in good agreement with the analytic solution for the critical value of τ_n in equation (73). They

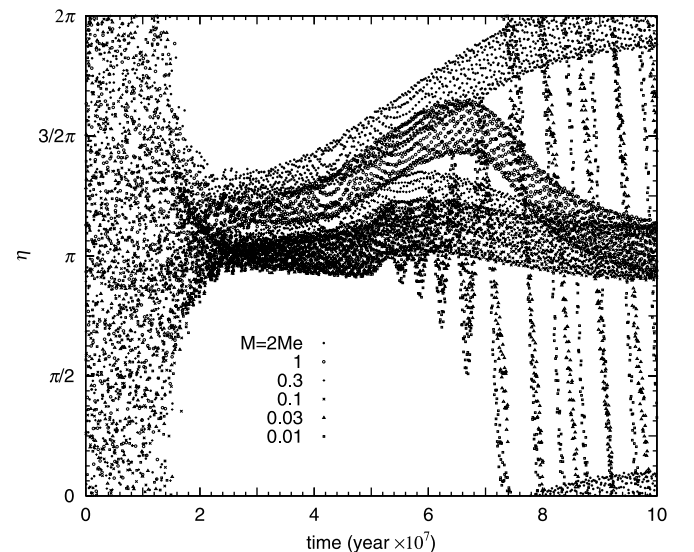


FIG. 6.—Relative angle of perihelion, $\eta = \varpi_p - \varpi_J$. The initial conditions are identical to those in Fig. 4b. Filled circles, open circles, plus signs, crosses, filled triangles, and filled squares show $m = 2, 1, 0.3, 0.1, 0.03$, and $0.01 M_\oplus$, respectively.

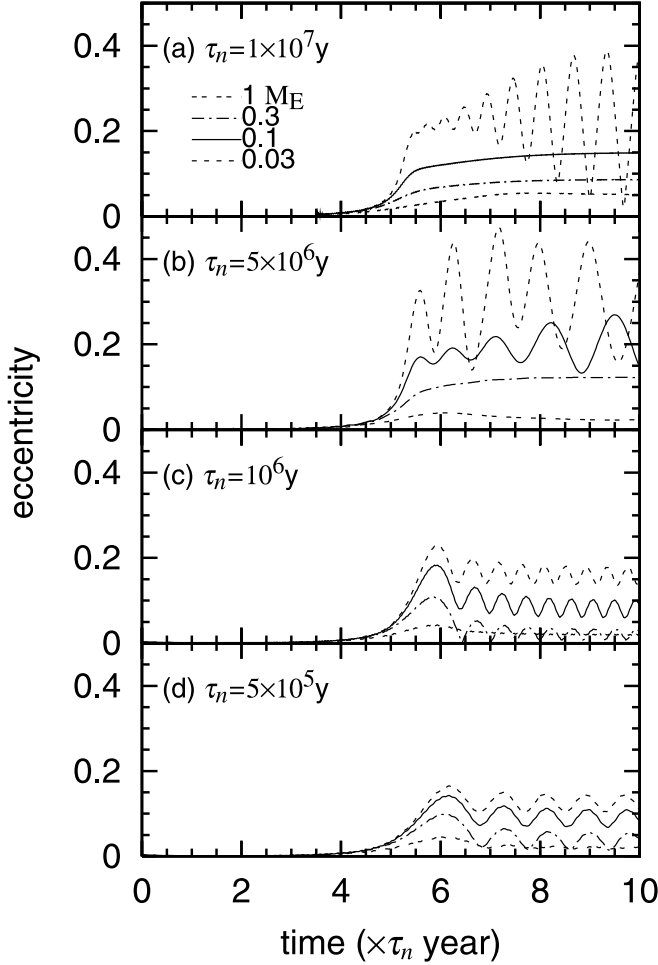


FIG. 7.—Eccentricity evolution dependence on the disk depletion timescale τ_n . Time is scaled by τ_n . The dashed, solid, dot-dashed, and double-dashed lines are $m = 0.03, 0.1, 0.3$, and $1 M_\oplus$, respectively. The initial semimajor axes are 1 AU. We use current the orbital parameters of the Jovian planets for their initial conditions.

suggest that capture by secular resonance is favorable for large τ_n , M_J , and e_J . For the solar system configuration, there is a strong tendency for the migration of resonant embryos to stall when they reach ~ 1 AU.

5.6. Equilibrium and Asymptotic Eccentricities

We now consider the dependence of the equilibrium eccentricity on different parameters. In Figure 8 we consider two sets of models with $e_J = 0.01$ and 0.045 , respectively. We use seven values of m_e ($0.01, 0.02, 0.03, 0.1, 0.3, 1$, and $2 M_\oplus$) and six values of m_e ($0.01, 0.03, 0.1, 0.3, 1$, and $2 M_\oplus$) for the first and second sets, respectively. We adopt five initial semimajor axes ($a_{\text{init}} = 1, 1.5, 2, 2.5$, and 3 AU). The final eccentricity is averaged after the ν_5 resonance has essentially stalled.

The results of the second ($e_J = 0.05$) set clearly indicate that eccentricities are independent of a_{init} . All the embryos are captured by the sweeping secular resonances, and they all attain an equilibrium eccentricity in which excitation induced by the secular resonance is balanced by the eccentricity damping. The e magnitude and its m_e dependence $e \propto m_e^{-1/3}$ are in general agreement with those derived analytically (see eq. [71]).

Among the first set of results, the resonance strength is too weak to capture the embryos. Consequently, they do not undergo any significant migration. Their eccentricity is briefly excited

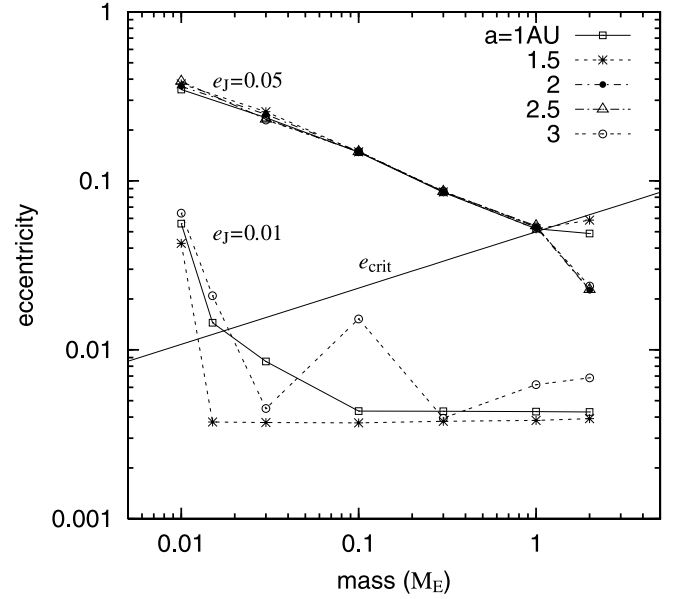


FIG. 8.—Final equilibrium eccentricities for two different Jovian eccentricities ($e_J = 0.045$ and 0.01). The values are averaged after the ν_5 has stalled. Initial semimajor axes are 1 (squares), 1.5 (asterisks), 2 (filled circles), 2.5 (triangles), and 3 AU (open circles). The line of e_{crit} represents the eccentricity at which orbit crossing occurs, assuming an embryo orbital spacing of $10r_H$, i.e., $e_{\text{crit}} = 10r_H/a \propto m^{1/3}$.

during the passage of the resonance by an amount comparable to those in equation (52). After the passage of the resonance, the embryos' eccentricities are damped efficiently, especially for those with relatively large masses. Since the damping timescales of the low-mass embryos are long, finite eccentricities are preserved. The m_e dependence ($e \propto m_e^{-1}$ in eq. [52]) is in good agreement with the numerical results.

For comparison purposes, we also plot in Figure 8 the embryos' critical eccentricity,

$$e_{\text{crit}} = \frac{\Delta a}{2a} \simeq 5 \left(\frac{m_e}{3M_*} \right)^{1/3} = 0.05 \left(\frac{m_e}{M_\oplus} \right)^{1/3} \left(\frac{M_*}{M_\odot} \right)^{-1/3}. \quad (31)$$

For $e > e_{\text{crit}}$, the embryos' radial excursion is larger than their feeding zone width. These results indicate that all embryos with mass less than $\sim 1 M_\oplus$ have a tendency to cross each other's orbits and to collide.

6. ORBITAL CROSSING OF PROTOPLANETS

Up to now, we have considered the response of isolated embryos to the propagation of sweeping secular resonances and tidal damping. In this section we consider the response of multiple embryos and demonstrate that the sweeping secular resonance can lead to enhanced orbit crossing and collisions.

6.1. Local Orbital Interaction

In order to establish the dependence of resonant trapping probability, we consider two sets of calculations. In the first set, we set $e_J = 0.01$. For illustration purposes, we place four embryos with isolation mass, which are separated by a full feeding zone width $\Delta a = 10r_H$, in the neighborhood of 1 AU. Using equation (4), we find (with $f_d = 0.7$) $M_{\text{iso}} \simeq 1/15 M_\oplus$. For comparison, we also consider a low disk mass case with much reduced isolation masses ($M_{\text{iso}} = 1/300 M_\oplus$, which corresponds to an order of magnitude less massive disk than a minimum-mass nebula).

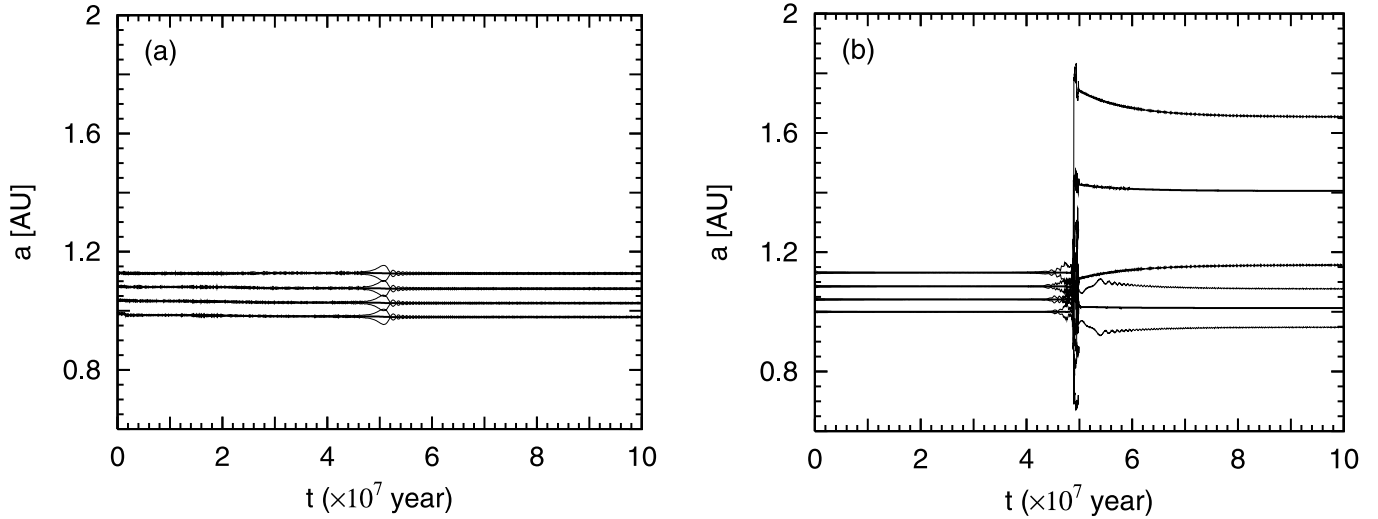


FIG. 9.—Orbital crossing of four isolation bodies. The semimajor axes and the perihelia and aphelia are shown by the thick and thin lines, respectively. Eccentricities of the Jovian planets are 0.01. The disk depletion timescale is chosen to be 10 Myr. (a) Isolation masses are 2×10^{-7} , 2.06×10^{-7} , 2.13×10^{-7} , and $2.19 \times 10^{-7} M_{\odot}$. They are initially separated by $10r_H$. The initial semimajor axes are about 1, 1.04, 1.09, and 1.13 AU. (b) Masses are $10^{-8} M_{\odot}$. The initial semimajor axes are the same as in (a).

The eccentricity damping timescale decreases with m_e (eq. [26]). For the standard MMNM, M_{iso} is sufficiently large that the embryos' eccentricity is efficiently damped. The results of our numerical calculation in Figure 9a show that the trapping conditions are no longer satisfied. Although the embryos are not trapped by the sweeping secular resonances, their eccentricities are excited to ~ 0.02 during the resonant passage. For such a weak excitation, the embryos' e is less than or comparable to e_{crit} and the system remains stable. This result is in agreement with the analytic approximation in the following section (see eq. [52]).

For the low-mass disk, the trapping conditions become even more difficult to satisfy. But, with a reduced damping efficiency, the embryos attain relatively large eccentricities during the approach and passage of the sweeping secular resonance ($e_{peak} \propto m_e^{-1}$). Consequently, the embryos' eccentricities are excited to magnitudes greater than their e_{crit} . Through orbit crossing and close encounters, embryos attain large eccentricities that are not effectively damped. For illustration purposes, we also considered a case in which damping through embryo-disk interaction and the gravity of two Jovian planets are neglected altogether (Fig. 10). In this case, the embryos' orbits become unstable prior to the arrival of the sweeping secular resonance and once again their eccentricities become very large. In all three sets of models, we do not take into account the merger process among the colliding embryos and planetesimals.

With the current Jovian eccentricity ($e_J \sim 0.05$), the magnitude of peak eccentricity (see eq. [52]) exceeds the critical value e_{crit} for embryos with $m_e = M_{iso}$ of the minimum-mass nebula to cross each other's orbit. The numerical results in Figure 11 show that during the approach of the sweeping secular resonance, e grows until the embryos cross each other's orbit. In this model, we consider the possibility of cohesive collisions. With an assumed internal density of 3 g cm^{-3} , we compute the size of the embryos. We assume that all collisions lead to mergers. In this model, two embryo-embryo mergers occur. One residual embryo is captured by the gas giant's secular resonance as it sweeps inward. The eccentricity of the captured embryo is maintained to a modest equilibrium value (similar to those given in eqs. [71]

and [67]). In contrast, the other remaining embryo is dislodged from the resonances by encounters and orbital repulsions. Outside the resonance, the embryo's eccentricity is damped and its migration is terminated. This result indicates that collisions or close encounters can detach the embryos from their sweeping secular resonance, and the surviving embryos can be preserved with low eccentricity and modest semimajor axis.

Numerical results in this section clearly indicate that the equilibrium and asymptotic eccentricities of the embryos depend on the magnitude of the e_J , M_J , τ_m , and m_e . Prior to the arrival of the sweeping secular resonance, $m_e \simeq M_{iso}$. Both M_{iso} , their Δa , and e_{crit} are functions of the disk mass. In the context of the solar system, the main goal is to identify regions of the parameter space that allow the excitation of e and growth through coagulation induced by the sweeping secular resonance and the efficient damping of the mergers' eccentricity that leaves behind an orderly planetary system.

6.2. Evolution near the Asteroid Region

The numerical results presented above are most relevant to embryos near 1 AU. We now place four isolation-mass embryos in the asteroid belt, a region between 2.5 and 3 AU (Fig. 12). Since this region is close to Jupiter, the sweeping secular resonance ν_5 is initiated from this region (see Figs. 1 and 2). It also contains Jupiter's strongest inner mean motion resonances, which strongly perturb the embryos. As the outermost embryo rapidly gains eccentricity, it is also being circularized by its tidal interaction with the disk. These combined influences excite a rapid growth of embryos' eccentricity and induce their orbits to cross.

After some initial adjustment, the separation between the three remaining embryos is enlarged such that they return to dynamical separation, at least temporarily. On the timescale of several million years, the gas giant's sweeping secular resonance propagates inward. As the outermost embryo is captured onto the sweeping secular resonance, it attains an equilibrium eccentricity, maintains an equilibrium longitude of periapse relative to Jupiter's, and migrates inward along the path we have described in the previous section.

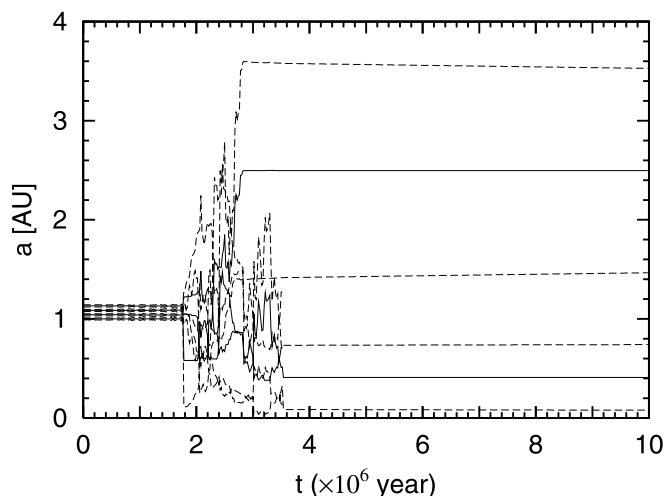


FIG. 10.—Orbital instability of four isolation embryos. Only the mutual gravity of the four embryos is included. The initial conditions of the embryos are the same as in Fig. 9a. The solid lines are semimajor axes, and the dashed lines are perihelia and aphelia.

During its inward migration, the outermost embryo approaches the orbit of its neighbor and undergoes close encounters at around 3 Myr and again at 8 and 18 Myr. But the strength of the sweeping secular resonance is sufficiently large that it keeps the outermost embryo locked. Asymptotically, most of the embryos merge and migrate to the location where the ν_5 sweeping secular resonance stalls, and the *asteroid belt region is cleared* (see Fig. 12).

6.3. Evolution Spanning a Wide Radial Range

Here we present some results of preliminary simulations of collisional evolution. In order to discuss the final spacing and final eccentricity of terrestrial planet analogs produced in this way, a systematic statistical analysis must be carried out. Such a survey will be presented in a companion paper (E. W. Thommes et al. 2005, in preparation). Here we present one model to illustrate some basic properties associated with the collisional evolution of a population of embryos as the gas giants' sec-

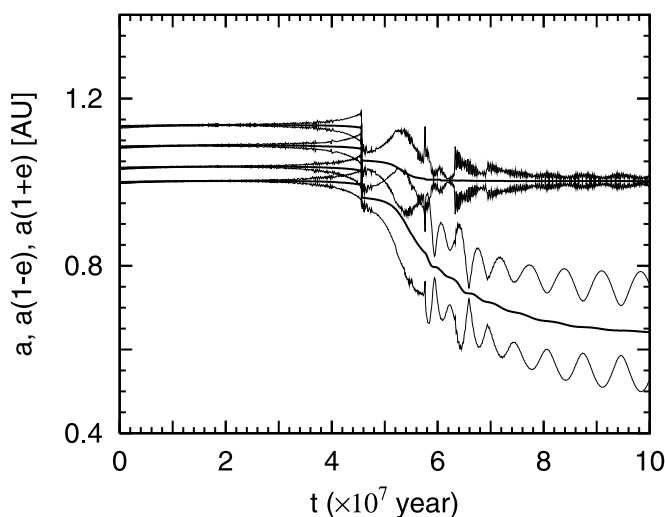


FIG. 11.—Orbital crossing of four isolation bodies with current Jovian eccentricity. The initial conditions are the same as in Fig. 9a, but the eccentricities of the Jovian planets are set to their present values.

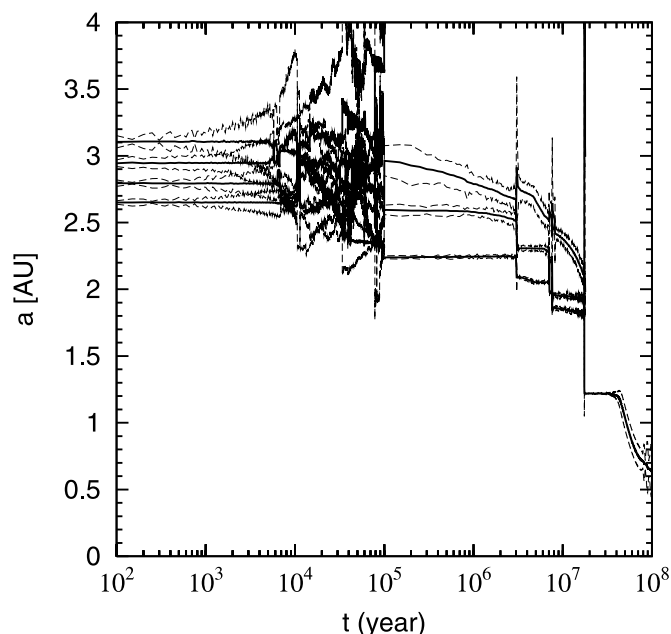


FIG. 12.—Orbital evolution of four isolation bodies in the asteroid region using current Jovian eccentricities. The solid lines show semimajor axes, and the dashed lines show perihelia and aphelia. The masses of the embryos are about 4.15×10^{-7} , 4.32×10^{-7} , 4.50×10^{-7} , and $4.69 \times 10^{-7} M_{\odot}$, and their initial semimajor axes are about 2.65, 2.79, 2.95, and 3.11 AU, respectively.

ular resonances sweep through the inner region of the solar system.

For our illustrative model, we adopt a surface density that is smaller than that of the minimum-mass nebula. We choose $f_d = 0.5$ such that the total mass of solid material within the inner 4 AU is comparable to the present-day total mass of the four terrestrial planets in the solar system. Within the range of 0.6–4 AU, we place 41 embryos, each with a local isolation mass M_{iso} and separated from their neighbors by $\Delta a = 13r_H$. For the gas giants, we use the present-day mass, eccentricity, and semimajor axis of Jupiter and Saturn. The initial mass of the gas in the disk is the same as that of the minimum-mass nebula with $f_g = 1$. The gas depletion timescale is chosen to be $\tau_n = 5$ Myr.

In Figure 13, the initial dynamical disturbances of the gas giants on the embryos are clearly visible, especially outside 2 AU. Under the influence of gas giants' secular and mean motion resonances, the eccentricities of embryos outside 2 AU are excited. After some orbit crossing and cohesive collisions within the first 0.1 Myr, the residual embryos attain dynamically separated orbits with relatively small eccentricities. Dynamical stability of embryos inside 2 AU is preserved until the sweeping secular resonance reaches them.

During the propagation of the sweeping secular resonance, the outermost embryos are captured onto it. They attain equilibrium eccentricities and maintain coherent longitude of periapse. Consequently, they undergo collisions with enhanced frequency (see eq. [77]). These collisions lead to the growth of m_e . Interior to 1.5 AU where the strength of the secular resonance is severely weakened, some collisions lead to the dislodging of the merger products out of the secular resonance. Outside the resonance, there is still adequate gas to damp the eccentricity. Eventually, four planets are left with semimajor axes at 0.5, 0.83, 1.27, and 1.65 AU and $e = 0.01, 0.01, 0.015$, and 0.016 , respectively. Since the retention efficiency of the building blocks is nearly unity, these emerging planets have masses comparable to the terrestrial planets.

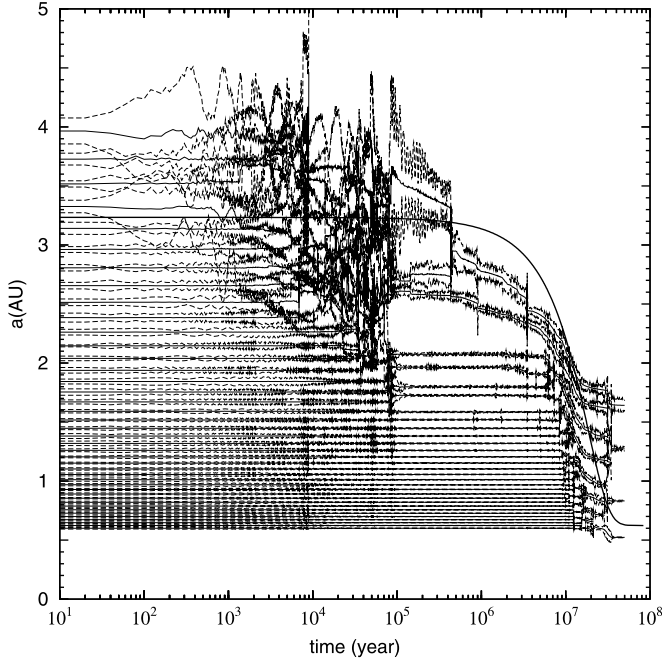


FIG. 13.—Collisional evolution of 41 embryos. Embryos are initially separated by $13r_H$. The disk depletion timescale is $\tau_n = 5$ Myr. The solid, dashed, and thick solid lines show semimajor axes, perihelia and aphelia, and the location of ν_5 resonance, respectively.

The final outcome is a stable (at least on the timescale of 100 Myr) system of Earth-mass planets with relatively low eccentricities. It can be applied in the context of solar system formation (Lin et al. 2005). The asteroid region is devoid of material. Some late-stage collisions between Earth- and Mars-size embryos may have interesting implications for the formation of the Moon. Mixing in the radial direction may also lead to enrichment of water content in some terrestrial planets. Finally, the high retention efficiency justifies the basic assumption underlying the MMNM. These outcomes are necessarily statistical in nature. We will examine the dependence on the depletion timescale, f_d , τ_n , and other parameters in a subsequent paper.

7. ANALYTIC APPROXIMATION OF THE EVOLUTION EQUATIONS

In this section we present a linearized analysis and obtain solutions to interpret the results of numerical computation presented in the previous section. In this analysis we first consider the secular perturbation due to a single gas giant and neglect the evolution of its eccentricity e_J . We assume that an equilibrium eccentricity may be quasi-statically maintained for the gas giant with surrounding gaps by the balancing influence of its corotation and Lindblad resonances (Goldreich & Tremaine 1982; Goldreich & Sari 2003), as well as nonlinear dissipation (Papaloizou et al. 2001). In the presence of additional gas giants, both their longitude of periastron precesses and the amplitude of their eccentricity is modulated.

For a constant $e_J (\ll 1)$, the embryos' equations of secular evolution can be approximated to the lowest order in e (Murray & Dermott 1999). But, in both cases we have just presented above, the embryos' e reaches sufficiently large magnitude that a first-order approximation may no longer be valid. Nevertheless, we utilize the approximate analytic expression for illustration and discussion purposes only.

In the limit of small embryos' eccentricity, we assume that the gas drag damps the embryos' eccentricity without removing

their orbital angular momentum. Averaging the equation of motion over an embryo's orbit, it is reduced (Mardling & Lin 2002) to a set of secular evolution equations,

$$\frac{de}{dt} = -\mu_J \sin(\varpi - g_J t - \lambda_J) - \frac{eu}{\tau_{\text{damp}}}, \quad (32)$$

$$\frac{d\varpi}{dt} = g_a - \frac{\mu_J}{e} \cos(\varpi - g_J t - \lambda_J), \quad (33)$$

$$\frac{da}{dt} = -a \left(\frac{2e^2}{1-e^2} \right) \frac{u}{\tau_{\text{damp}}}, \quad (34)$$

where

$$u = \exp\left(-\frac{t}{\tau_n}\right), \quad (35)$$

$$\mu_J = \frac{|C_{p,J}|}{t_{p,J}} e_J, \quad (36)$$

and λ_J is a phase angle of Jupiter at $t = 0$. The subscript a refers to variables associated with embryos at a heliocentric distance a , and we substitute it for p in equation (15). The expressions for g_a and g_J can be obtained from equations (10), (14), (15), and (17). Substituting

$$H \equiv e \sin \eta, \quad K \equiv e \cos \eta \quad (37)$$

for $e = (H^2 + K^2)^{1/2}$ and the relative longitudes of the residual embryos and Jupiter,

$$\eta = \tan^{-1}\left(\frac{H}{K}\right) \equiv (\varpi - g_J t - \lambda_J), \quad (38)$$

equations (32) and (33) reduce to

$$\frac{dH}{dt} = \mu_J - \frac{u}{\tau_{\text{damp}}} H + (g_a - g_J) K, \quad (39)$$

$$\frac{dK}{dt} = -\frac{u}{\tau_{\text{damp}}} K - (g_a - g_J) H. \quad (40)$$

7.1. Two Illustrative Models

Two sets of numerical solutions of equations (32)–(34) for the eccentricity evolution through the passage of the secular resonance are shown in Figure 14. In the first case we consider the limit of no damping by setting $\tau_{\text{damp}} = \infty$, whereas in the second case we approximate the damping effect with equation (8). We used as initial orbital parameters a semimajor axis of 1 AU, $\tau_n = 10$ Myr, and $m_e = 0.03 M_{\oplus}$. The eccentricity can be greatly excited at the time of the secular resonance passage ($\tau \sim 5$ Myr).

In the absence of damping, the embryo's semimajor axis does not evolve and the eccentricity increases rapidly. The above discussion indicates that the embryos' secular response becomes circulation rather than libration when the eccentricity increases to where $\mu_J/e < g_a$. The asymptotic eccentricity is 0.8. For such large eccentricity, the above equations and the linear analysis of secular resonance in § 3 (i.e., the expression in eq. [9]) are no longer valid. We find that any further linear eccentricity excitation is quenched by higher order terms.

But, with the damping, the eccentricity only reaches 0.6 during the passage of the secular resonance at $t_0 \approx 55$ Myr. In this

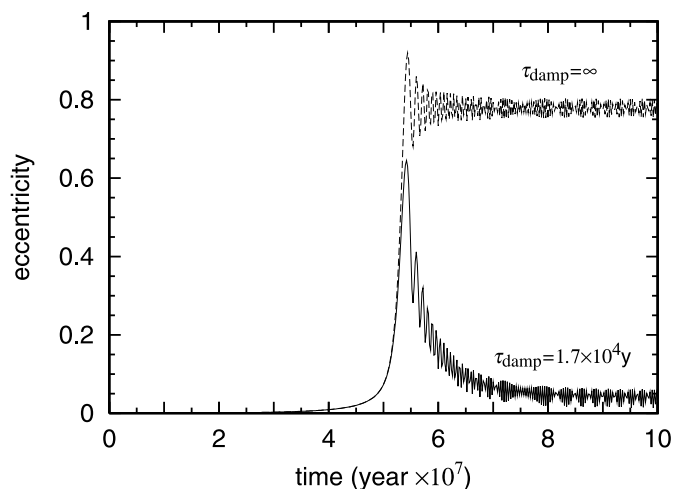


FIG. 14.—Numerical solutions of equations (32) and (33) for the eccentricity evolution of a $0.03 M_{\oplus}$ embryo at 1 AU. A damping timescale ($\tau_{\text{damp}} = 1.7 \times 10^4$ yr) evaluated from equation (8) is adopted in the solution shown by the solid line. The dashed line is a case without tidal damping. The disk depletion timescale is $\tau_n = 10$ Myr in both cases.

idealized calculation, we neglected equation (34) and, without any embryo semimajor axis evolution, the sweeping resonance passes through the embryo's orbit. Thereafter, the asymptotic rate of net eccentricity excitation declines, essentially following the path given by equation (28), and the embryo's e is reduced to 0.05 at 100 Myr.

In Figures 15a and 15b we consider another series of models to illustrate the induced evolution of the embryos' semimajor axis. When their eccentricity is excited, the tidal drag due to embryo-disk interaction intensifies. The damping of their eccentricity is accompanied by the decrease of their semimajor axes. In our integrations, the rate of semimajor axis evolution is directly determined by equation (34). We consider a range of embryo masses from 0.01 to $10 M_{\oplus}$. Here we also adopt $a = 1$ AU, $\tau_n = 10$ Myr, and $e_J = 0.05$. At the onset of the calculation,

embryos with masses larger than the isolation mass ($\sim 1/15 M_{\oplus}$) attain very small eccentricities and already suffer a small amount of orbital decay. But, in all cases, the eccentricity of the embryos is greatly excited, and they migrate inward substantially after the secular resonance has swept past their initial semimajor axis.

With the exception of the $m_e = 10 M_{\oplus}$ case, the embryos migrate at the appropriate pace to maintain a state of marginal secular resonances. This state is attained through the self-adjustment of the embryos' eccentricities. We note that the migration rate is insensitive to m_e . Although $\tau_{\text{damp}} \propto m_e^{-1}$ (see eq. [8]), the equilibrium eccentricity also decreases with m_e . Small embryos damp relatively slowly but attain large eccentricities, whereas the eccentricities of large embryos damp rapidly and attain small or modest eccentricities. Consequently, there is no significant difference in the final outcome between embryos of isolation mass and embryos of an Earth mass. However, the critical disk mass that induces the onset of the resonant migration is a function of τ_n and e_J . For the most massive ($10 M_{\oplus}$) case, the damping efficiency is so high that only a small equilibrium eccentricity can be maintained. The resulting orbital decay cannot keep pace with the propagation of the secular resonance.

7.2. Initial Evolution

The results in Figures 15a and 15b indicate that prior to the passage of the secular resonance, the disk potential dominates the precession of both Jupiter and the embryos. The embryos' eccentricity is modulated with a small amplitude. Since the secular interaction does not induce energy exchange between gas giants and embryos to first order, the change of the embryos' a directly associated with it is negligible. The embryos also interact with the disk. If the tidal torque due to embryos' interaction with both sides of the disk is balanced, there is also no net exchange of angular momentum between the embryos and the disk. Nevertheless, the eccentricity damping due to the embryo-disk tidal interaction can lead to orbital evolution. In the above expressions, the change of the semimajor axis is only introduced by this damping process.

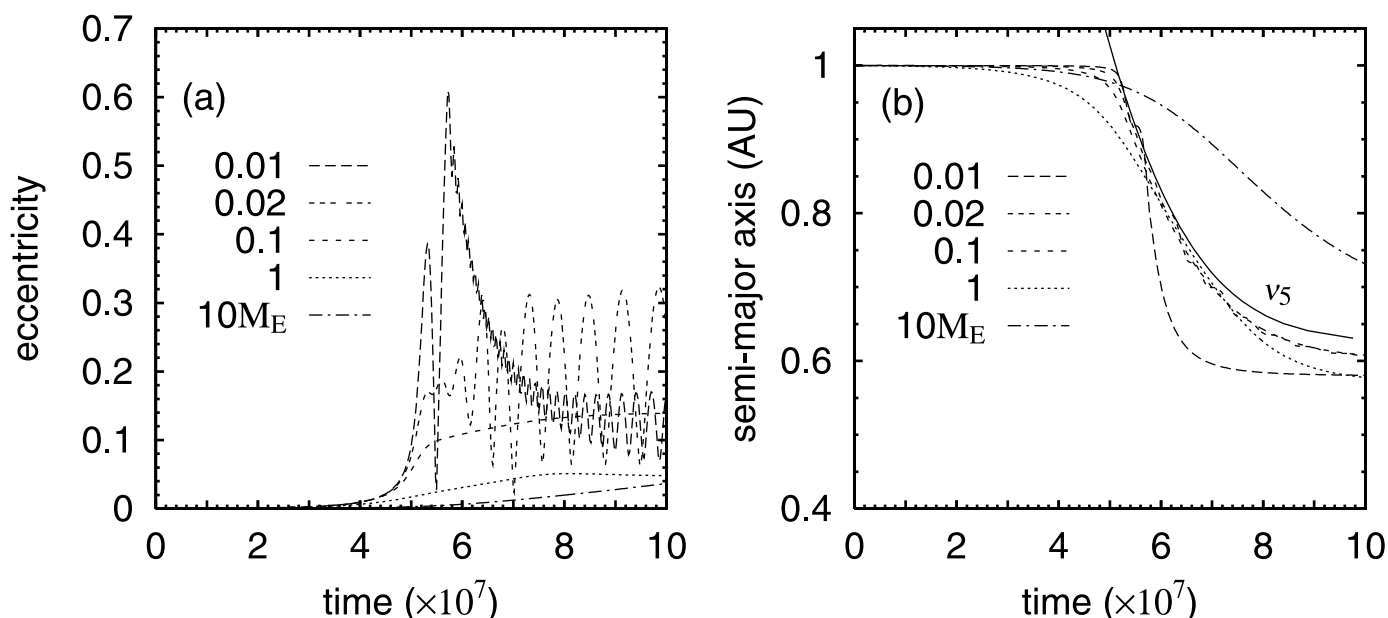


FIG. 15.—Numerical solutions of equations (32), (33), and (34). The long-dashed, triple-dashed, short-dashed, dotted, and dot-dashed lines are the evolution of (a) eccentricities and (b) semimajor axes of $m = 0.01, 0.02, 0.1, 1$, and $10 M_{\oplus}$ embryos, respectively. Embryos are initially at 1 AU, and the disk depletion timescale is $\tau_n = 10$ Myr. The Jovian eccentricity is 0.05. The solid line in panel (b) shows the location of the ν_5 resonance.

This qualitative description can be demonstrated with a quantitative analysis. Using the substitutions

$$\Delta g \equiv g_a - g_J, \quad (41)$$

$$b \equiv \frac{u}{\tau_{\text{damp}} \Delta g} = \left[\frac{1}{g_{a,J} + (g_{a,\text{disk}} - g_{J,\text{disk}})u} \right] \frac{u}{\tau_{\text{damp}}}, \quad (42)$$

$\phi \equiv \int \Delta g dt$, and $D \equiv d/d\phi$, the approximate linearized equations (39) and (40) can be reduced to

$$(D + b)H - K = \frac{\mu_J}{\Delta g}, \quad (43)$$

$$H + (D + b)K = 0. \quad (44)$$

If we neglect the radial migration, b is approximately constant and the solutions of the above equations become

$$H = \exp(-b\phi)[(C_1 + c) \cos \phi + (S_1 + s) \sin \phi], \quad (45)$$

$$K = \exp(-b\phi)[-(C_1 + c) \sin \phi + (S_1 + s) \cos \phi], \quad (46)$$

where c and s are constants of the integral and

$$C_1 = \int \cos \phi \frac{\mu_J}{\Delta g} \exp(b\phi) d\phi, \quad (47)$$

$$S_1 = \int \sin \phi \frac{\mu_J}{\Delta g} \exp(b\phi) d\phi. \quad (48)$$

Outside the resonance where the magnitude of Δg is non-vanishing, the amplitudes of C_1 and S_1 are bounded, whereas that of H and K decays with u . But in resonances, C_1 and S_1 approach singularity and e can be excited to modest values.

7.3. Passage of the Secular Resonance

We define $t = t_R$ to be the instance when the center of the secular resonance (where $\Delta g = 0$) passes through the initial location of the embryos. From equations (15), (17), and (41) we obtain

$$t_R = \tau_n \ln \left(\frac{g_{J,\text{disk}} - g_{a,\text{disk}}}{g_{a,J} - g_{J,S}} \right). \quad (49)$$

For $t > t_R$, the resonant location migrates inward from the embryos' initial a (also see Fig. 2). In the above equation, $g_{a,\text{disk}}$ is always negative (see eq. [13]). In § 3.4 we show that if a gas giant planet has acquired a sufficiently large mass to open a gap in the disk, $g_{J,\text{disk}} > 0$ (see eq. [23]). If it cannot form a gap, $g_{J,\text{disk}} < 0$ and has a magnitude $< |g_{a,\text{disk}}|$ for the embryos with $a < a_J$. In either case, the right-hand side of equation (49) is well defined and the secular resonance can be attained (also see the numerical results in Fig. 2 and discussions in § 3.4). The magnitude of t_R deduced from equation (49) is in good agreement with that obtained in the full numerical integration (see Fig. 3). Note that in the absence of Saturn, $g_{J,S} = 0$. In general, Saturn's contribution limits the radial extent over which the secular resonance may sweep through the solar system.

7.3.1. The Absence of Eccentricity Damping

With a substitution, $t' \equiv t - t_R$, we can expand, in the limit $t' \ll t_R$ and near the secular resonance, $\exp(-t'/\tau_n) \sim 1 - t'/\tau_n$

and $u = t_R(1 - t'/\tau_n)$. The difference in the precession rates can be expressed as $\Delta \dot{g} = \Delta \dot{g}_d t'$, where $\Delta \dot{g}_d = d\Delta g/dt'$. In the absence of any damping, $\dot{a} = 0$, $b = 0$, and C_1 and S_1 become Fresnel integrals. Provided that the initial e is small, a remains constant and e reaches

$$e_{\text{max}} \sim \left(\frac{2\pi}{\Delta \dot{g}_d} \right)^{1/2} \mu_J. \quad (50)$$

In this limit, the only time dependence in g_a , g_J , and Δg is through u , and the propagation speed of the resonance is inversely proportional to the disk depletion timescale, i.e.,

$$\Delta \dot{g}_d \simeq \dot{g}_{a,\text{disk}} - \dot{g}_{J,\text{disk}} \sim \frac{(g_{a,\text{disk}} - g_{J,\text{disk}})}{\tau_n}. \quad (51)$$

The maximum eccentricity increases in proportion to $\tau_n^{1/2} e_J$. This conclusion is in agreement with the numerical results (see Figs. 4 and 14).

7.3.2. The Limit of Inefficient Eccentricity Damping

Finite values of b induce two effects: (1) the damping of eccentricity that is included in equations (39) and (40), and (2) orbital decay (i.e., a decrease in a) through equation (34). Although τ_{damp} evolves with a , the dominant effects of orbital migration are in the continual adjustment of both g_a and g_J . We consider here these two effects separately.

In the limit of small but finite damping rate or, equivalently, large τ_{damp} , $b\phi = ut'/2\tau_{\text{damp}} \ll 1$ during the approach and shortly after the passage of the sweeping secular resonance, while a does not change significantly. The embryos' e reaches its first peak value

$$e_{\text{peak}} \sim \tau_{\text{damp}} \mu_J \exp\left(\frac{t_R}{\tau_n}\right) \sim 0.03 \left(\frac{m_e}{m_{\oplus}}\right)^{-1} \left(\frac{a}{1 \text{ AU}}\right)^{-3} \left(\frac{M_J}{10^{-3} M_{\odot}}\right) \left(\frac{e_J}{0.05}\right) \quad (52)$$

when de/dt (in eq. [32]) first vanishes after the passage of the resonance (see Fig. 4a). In the above estimation, we adopt the current semimajor axis and eccentricity values of Jupiter. We also assume η to be close to $\pi/2$ or $3\pi/2$. If η has values close to 0 or π , the magnitude of e_{peak} would be reduced (see Figs. 6 and 14).

7.3.3. Avoidance of Resonant Trapping

In the low- b limit, the decline of the semimajor axis cannot keep pace with the sweeping rate of secular resonance. After the passage of the resonance, the embryos' eccentricity decreases exponentially due to the weak tidal drag such that

$$e = e_{\text{max}} \exp \left\{ \frac{\tau_n}{\tau_{\text{damp}}} \left[\exp\left(-\frac{t}{\tau_n}\right) - \exp\left(-\frac{t_R}{\tau_n}\right) \right] \right\}. \quad (53)$$

After the depletion of the disk (i.e., $t' > \tau_n$), the embryos' e attains a finite asymptotic value

$$e_{\text{asym}} = e_{\text{max}} \exp \left[-\frac{\tau_n}{\tau_{\text{damp}}} \exp\left(-\frac{t_R}{\tau_n}\right) \right]. \quad (54)$$

This nonvanishing asymptotic e_{asym} is attained because the disk gas is depleted before the embryos' e is completely damped (see Fig. 4a).

7.4. Protracted Secular Resonant Trapping of Embryos

We now consider moderate damping rates, in which case the embryos' a evolves during the passage of the secular resonance. Residual embryos are trapped in the sweeping secular resonance when their induced orbital decay enables their precession frequency g_a to evolve with that of the gas giant g_J . For illustration purposes, we consider the influence of only one giant planet and utilize the linearized secular equations (39) and (40) even though the embryos' eccentricities may reach sufficiently large values to invalidate these approximations. The approximate solutions are used to interpret the numerical results based on integration of the full equations of motion in the previous sections.

7.4.1. Resonance Capture Conditions

Numerical results indicate that embryos migrate with the sweeping secular resonance. During the decline of their semi-major axes, nearly constant values of e are sustained (see Figs. 4b and 5). There are four conditions for establishing this quasi-equilibrium state in which secular resonance may be sustained:

$$DH \simeq 0, \quad DK \simeq 0, \quad \Delta g \simeq 0, \quad \Delta \dot{g}_d \simeq 0. \quad (55)$$

From equations (43) and (44), we find that the first two conditions are satisfied when

$$H = \left(\frac{\mu_J}{\Delta g} \right) \left(\frac{b}{1 + b^2} \right), \quad (56)$$

$$K = -\frac{H}{b}. \quad (57)$$

From equation (37), $\eta = \tan^{-1}(K/H)$. In a quasi-equilibrium state, $\eta \sim \tan^{-1}(-b^{-1})$ such that $e \sim |H|$ and $\eta \sim \pi$ for $|b| \gg 1$ and $e \sim |K|$ and $\eta \sim \pi/2$ for $|b| \ll 1$.

The slowly varying H and K in equation (55) can also be derived directly from equations (32) and (33). With the definition of η in equation (38), we find from equation (33)

$$\frac{d\eta}{dt} = \Delta g - \frac{\mu_J}{e} \cos \eta. \quad (58)$$

Initially, $\Delta g < 0$ and $|\Delta g| \gg \mu_J/e$ so that the embryos attain circulating orbits. During the disk depletion, $|\Delta g|$ decreases and a transition from circulation to libration occurs when it is possible for $d\eta/dt \simeq 0$. From the above equation, this quasi-equilibrium is established first at $\eta = \pi$ as we have indicated in the previous paragraph for the large $|b|$ (or small $|\Delta g|$) limit. With a small increase in η , e can also attain an equilibrium value (see § 7.4.2). The equilibrium of eccentricity is only possible with negative $H \propto \sin \eta < 0$ (eq. [32]). Equation (58) also indicates that the equilibrium condition can be maintained in the limit of vanishing Δg only for $\eta \simeq 3\pi/2$.

In the absence of any gas giant planet's migration, the fourth condition in equation (55) can be satisfied if the embryos can undergo orbital migration faster than the sweeping speed of the secular resonance. The timescales for the embryos' migration and for the propagation of the sweeping secular resonance are

given by equations (34) and (22), respectively. The condition for resonance trapping becomes

$$\left| \frac{a}{\dot{a}} \right|_{\min} \simeq \left(\frac{1 - e_{\max}^2}{2e_{\max}^2} \right) \frac{\tau_{\text{damp}}}{u} < \tau_{\text{res}} \simeq 4\tau_n. \quad (59)$$

In the above expression, the minimum migration timescale is achieved when the embryos attain their maximum eccentricity. These equations clearly indicate that the embryos must acquire a sufficiently large $e_{\max}$ in order for them to be trapped in the resonance. In systems with low M_J or e_J , the secular resonance does not induce a sufficiently large $e_{\max}$ (see eqs. [67] and [71]) for rapid migration, so that the resonant capture conditions cannot be satisfied. Although $e_{\max}$ also increases as $\tau_{\text{damp}}^{1/3}$, in systems with large τ_{damp} and relatively small τ_n , resonant capture also cannot occur.

The sweeping secular resonance of massive, eccentric planets, however, can induce a radial migration of the embryos. Once they are captured into the secular resonance, the embryos adjust their eccentricity to maintain a quasi-equilibrium in which all the conditions in equation (55) are satisfied. In a minimum-mass nebula, we find from equations (10), (13), and (41) that the propagation rate associated with this quasi-equilibrium, $\Delta \dot{g} \simeq 0$, is

$$\frac{\dot{a}_{\text{res}}}{a_{\text{res}}} \simeq \left(\frac{g_{J,\text{disk}} - g_{a,\text{disk}}}{\beta_J g_{a,J} + g_{a,\text{disk}} u} \right) \frac{u}{\tau_n}, \quad (60)$$

where $\beta_J \equiv d \ln t_{p,J} / d \ln a$. Since the Laplace coefficient $b_{3/2}^{(1)}(\alpha)$ in the expression of $t_{p,J}^{-1}$ is an increasing function of α , $\beta_J \sim -2$. Thus, equation (60) implies that the protracted maintenance of secular resonance requires the embryos to undergo an inward migration. Since the right-hand side of equation (60) is independent of m_e , the propagation rate is the same for all planetesimals (see Fig. 5).

Equation (34) indicates that embryos indeed migrate inward as a consequence of their eccentricity damping due to their tidal interaction with the disk. Combining it with equation (60), the condition for sustained secular resonance becomes

$$\left| \frac{g_{a,\text{disk}} - g_{J,\text{disk}}}{\beta_J g_{a,J} + g_{a,\text{disk}} u} \right| \leq \left(\frac{2e^2}{1 - e^2} \right) \frac{\tau_n}{\tau_{\text{damp}}}. \quad (61)$$

The above equation is a more rigorous derivation of equation (59).

7.4.2. Initial Evolution of the Trapped Embryos

Initially, the embryos' precession is dominated by the disk potential such that $|g_{a,\text{disk}}| \gg \beta_J g_{a,J}$. Prior to reaching the secular resonance, the embryos' e is small and the resonance capture condition in equation (61) cannot be satisfied. But when the magnitude of Δg declines below

$$\Delta g_{\text{crit}} = \mu_J/e, \quad (62)$$

it is possible for the right-hand side of equation (58) to vanish, first with $\eta = \pi$ and later with larger values, $\eta = \pi + \delta\eta$. Thereafter, η attains a quasi-stationary phase angle such that e can grow substantially (see eq. [32]). For sufficiently large e , the right-hand side of equation (32) vanishes when

$$\delta\eta \simeq \frac{eu}{\mu_J \tau_{\text{damp}}} \quad (63)$$

(see Fig. 6). The magnitude of $\delta\eta$ is a decreasing function of both M_J and e_J ; i.e., a strong Jovian secular perturbation promotes the alignment of the embryo's periape with that of Jupiter.

When the sweeping secular resonance approaches an embryo, its $|\Delta g| < |g_{a,J}|$. But, after passage of the resonance, the eccentricity of the captured embryo adjusts so that its orbital decay and the resonance propagation timescales are equal. During this phase, provided that

$$\frac{2e^2}{(1-e^2)} \frac{\tau_n}{\tau_{\text{damp}}} > \left| \frac{g_{a,J}}{g_{a,\text{disk}} u^2} \right|, \quad (64)$$

we find, from equation (61), that

$$\Delta g \simeq \left(\frac{2e^2}{1-e^2} \right) \frac{\tau_n}{\tau_{\text{damp}}} g_{a,\text{disk}} u^2, \quad (65)$$

which is negative. The above equation indicates that it is possible to maintain the resonance condition with the magnitude of Δg being greater than that of $g_{a,J}$. We also find, from equation (42), that

$$b \simeq \frac{1-e^2}{2e^2 \tau_n g_{a,\text{disk}} u}, \quad (66)$$

which is likely to be large and negative (i.e., $\ll -1$) for small e . We have already indicated in equation (57) that large and negative b corresponds to $\eta \sim \pi$, which is independent of the embryos' mass.

7.4.3. Quasi-Equilibrium Evolution

Shortly after the resonance capture, the embryos' $\Delta g \simeq \Delta g_{\text{crit}}$. From equations (62) and (65), the embryos attain an equilibrium eccentricity for which

$$\frac{e_{\text{equi}}^3}{1-e_{\text{equi}}^2} \simeq \frac{\mu_J \tau_{\text{damp}}}{2\tau_n g_{a,\text{disk}} u^2}. \quad (67)$$

The magnitude of $e_{\text{equi}} \propto e_J^{1/3} m_e^{-1/3}$ and is insensitive to time. This equation agrees well with the numerical results in § 5 (see Figs. 4b and 8). For this equilibrium e , we find from equation (34) that the migration timescale $\dot{a}/a \propto M_J^{2/3} e_J^{2/3} m_e^{1/3} u^{-1/3}$. For low-mass embryos perturbed by a gas giant planet with the present-day values of M_J and e_J , the approximation in equation (64) is satisfied self-consistently. For high-mass embryos, more complex expressions for b , η , and e can be obtained by taking into account the added effect of their perturbation on the gas giant.

From equations (59) and (67), we estimate the critical condition for initially trapping the embryos by secular resonance to be

$$\tau_n > \frac{\tau_n u}{128} \left(\frac{g_{a,\text{disk}}}{\mu_J} \right)^2 \sim 240 e_J^{-2} \left(\frac{a}{1 \text{ AU}} \right)^{-1} \left(\frac{m_e}{M_{\oplus}} \right)^{-1} \text{ yr}. \quad (68)$$

This condition is difficult to satisfy for small e_J .

Provided that the above condition is satisfied, embryos migrate with the propagating secular resonance. As the disk depletes, the embryos' precession becomes mostly induced by the secular perturbation of the gas giant (i.e., $g_{a,J} > |g_{a,\text{disk}} u|$).

Provided that the condition in equation (64) is satisfied, equation (61) reduces to

$$\Delta g \simeq \left(\frac{2e^2}{1-e^2} \right) \frac{\tau_n g_{a,J}}{\tau_{\text{damp}}} \beta_J u, \quad (69)$$

which is positive. From equation (42), it follows that

$$b \simeq \left(\frac{1-e^2}{2e^2} \right) \frac{t_{a,J}}{\tau_n \beta_J}, \quad (70)$$

which declines with time such that η gradually decreases to $\pi/2$ from π . In principle, the asymptotic eccentricity in this limit is $\sim |K|$, and it can be determined from

$$\begin{aligned} \frac{e_{\text{asym}}^3}{1-e_{\text{asym}}^2} &\simeq \frac{\mu_J \tau_{\text{damp}}}{2|\beta_J| g_{a,J} \tau_n u} \\ &\sim 2 \times 10^{-4} \left(\frac{e_J}{0.05} \right) \left(\frac{m_e}{M_{\oplus}} \right)^{-1} \left(\frac{a}{1 \text{ AU}} \right)^{-1} \left(\frac{\tau_n}{1 \text{ Myr}} \right)^{-1}, \end{aligned} \quad (71)$$

which implies

$$e_{\text{asym}} \sim 0.06 \left(\frac{m_e}{M_{\oplus}} \right)^{-1/3}. \quad (72)$$

The above e_{asym} asymptotically reaches a finite value (also see Figs. 4b and 8). During the advanced stages of disk depletion, the sweeping rate of the secular resonance diminishes and therefore smaller eccentricity is needed for the embryos to acquire a matching migration rate. Numerical results indicate that the magnitude of e is a weakly growing function of time (see Fig. 4b). They represent a transition between equations (67) and (71) rather than the limiting case that is represented by equation (71).

The equilibrium values of e are maintained so long as the embryos are trapped in a secular resonance. From equations (59) and (71), we estimate the self-consistent critical condition for sustaining secular resonance trapping to be

$$\tau_n > \frac{\beta_J^2 g_{a,J}^2 \tau_{\text{damp}}}{128 u \mu_J^2} \sim 26 \left(\frac{m_e}{M_{\oplus}} \right)^{-1} \left(\frac{a}{1 \text{ AU}} \right)^{-4} \left(\frac{e_J}{0.05} \right)^{-2} \text{ Myr}. \quad (73)$$

In the above equation, we assume $|\beta_J| g_{a,J} \sim 200(a/1 \text{ AU})^{-1}(e_J/0.05)^{-1} \mu_J$ and utilized equations (21) and (26). Equation (73) indicates that the critical condition for the maintenance of secular resonance is a steeply decreasing function of a . The critical semimajor axis to maintain the secular resonance trapping,

$$a \sim 2 \left(\frac{\tau_n}{\text{Myr}} \right)^{-1/4} \left(\frac{m_e}{M_{\oplus}} \right)^{-1/4} \left(\frac{e_J}{0.05} \right)^{-1/2} \text{ AU}, \quad (74)$$

has a weak dependence on m_e as shown in Figures 5 and 7. Planetesimals located at small a are far away from Jupiter, and

by the time that the secular resonance has propagated to such locations, its strength is significantly weakened. During the final stages of the gas depletion, the necessary condition for resonant trapping in equation (73) may no longer be satisfied before the propagation of the ν_5 resonance is stalled. Collisions can also dislodge embryos from the resonances. In these cases, the embryos' eccentricity would damp on the timescale of τ_{damp}/u . But if the migration of the embryos stalled together with the propagation of the ν_5 resonance, the equilibrium value in equation (71) would correspond to a finite asymptotic eccentricity.

7.4.4. Enhanced Collision Frequency

The results in equation (71) indicate that the quasi-equilibrium eccentricity of the resonant embryos can be excited above $\Delta a/a$ such that the extent of their radial excursion surpasses the width of the feeding zone (see Fig. 8). Although the resonant excitation of their eccentricity enables the embryos to cross each other's orbit and to collide, their elevated velocity dispersion σ may also lead to a reduction in their collision frequency and growth rate (see eq. [1]).

However, the embryos' velocity dispersion may be suppressed by the relatively weak m_e dependence of the magnitude of e and η , which implies that the orbits of embryos with a wide range of masses are nearly aligned. With their $a \sim a_{\text{res}}$, the resonant embryos migrate together with small dispersions in e ($\delta e < e$). Since the evolution of the resonance location is determined by the disk depletion time only (see eq. [60]), the resonant embryos migrate together. The near alignment of η suppresses the dynamical instability of the systems. But with a given dispersion in the embryo's mass δm_e , a small dispersion in e is expected. In the limit that $\delta m_e < m_e$, $\delta e = \beta_m \delta m_e$, where $\beta_m = d \ln e / d \ln m_e \sim -\frac{1}{3}$ during the quasi-equilibrium evolutionary phase (see eq. [71]). From equations (63) and (71) we find

$$\delta \eta \sim \left(\frac{u}{\mu_J \tau_{\text{damp}}} \right)^{2/3} \left(\frac{1}{2\tau_n \beta_J g_{a,J}} \right)^{1/3} \propto m_e^{2/3} \quad (75)$$

(also see Fig. 6). For a limited dispersion $\delta m_e (< m_e)$, $\delta e/e \sim -\frac{1}{3} \delta m_e/m_e$ such that $\delta \eta < \delta e/e$ especially during the late phases when $u \ll 1$. Despite this apsidal alignment, the resonant embryo's orbits cross each other provided that their $\delta e > \delta \ln a$. As we have already indicated that a_{res} depends only on the global depletion of the disk, the magnitude of δa is expected to be negligibly small and the condition for orbit cross is satisfied. The velocity dispersion $\sigma \sim \delta e \Omega_K a_{\text{res}} < e \Omega_K a_{\text{res}}$ among a population of embryos is

$$\sigma \simeq \left(\frac{\delta m_e}{m_e} \right) \beta_m e \Omega_K a_{\text{res}}. \quad (76)$$

From equation (1), we estimate the growth timescale to be

$$\tau_{\text{growth}} \sim \left(\frac{\beta_m \delta m_e}{2\pi m_e} \right)^2 \left(\frac{a}{R} \right) \frac{M_*}{M_{\text{res}}} P, \quad (77)$$

where M_{res} is the total mass of embryos trapped in the resonance. From equations (76) and (77), we estimate $\tau_{\text{grow}} \sim (1-10)(M_{\oplus}/m_e)(M_{\oplus}/M_{\text{res}})$ Myr.

In § 6 and a companion paper, we present numerical results to show that while the excited eccentricities enable the resonant

embryos to collide, the near alignment of their longitudes of periape reduces the relative speeds, enlarges their collisional cross sections, and promotes merging collisions between them (see Fig. 13).

7.4.5. Detachment from Sweeping Secular Resonance

Embryos exit the sweeping secular resonances via two avenues. Collisions and encounters of planetesimals introduce a finite probability for dislodging the trapped embryos out of the resonance. Cohesive collisions lead to a new set of semimajor axis (a') and eccentricity (e'). Outside the secular resonances, the embryos' e' generally decays on a timescale faster than τ_n . In the limit that their tidal interaction with their nascent disks does not significantly modify the embryos' angular momentum, the asymptotic semimajor axis is $a'' = a'(1 - e'^2)$. If a'' is outside the instantaneous a_{res} , the merger products are left behind the sweeping secular resonance (see Fig. 11).

In the absence of any collisions, embryos may also become detached from the migration path of the sweeping secular resonance when their nascent disks are severely depleted. At advanced stages of disk depletion, the condition in equation (64) is not satisfied. In this limit, equation (61) reduces to $\Delta g = g_{a,J}$ and $b \simeq u/(g_{a,J} \tau_{\text{damp}})$. In the asymptotic limit, b vanishes, and η and e approach $\sim \pi/2$ and $\sim |C_{p,J}|e_J$, respectively. But the trapping condition in equation (61) can no longer be sustained when u is reduced below

$$u_{\text{crit}} \simeq \left(\frac{1 - e^2}{2e^2} \right) \frac{\tau_{\text{damp}}}{\tau_n} \left| \frac{g_{J,\text{disk}} - g_{a,\text{disk}}}{g_{a,\text{disk}}} \right| - \beta_J \frac{g_{a,J}}{g_{a,\text{disk}}}. \quad (78)$$

For $t > \tau_n \ln u_{\text{crit}}$, the differential disk potential-induced precession between the embryos and the gas giant can no longer lead to a resonant condition in which Δg vanishes. Thereafter the propagation of the sweeping resonance is stalled and the embryos are no longer trapped by it.

8. CONCLUSIONS

In this paper we have studied the effect of disk depletion on the final assemblage of the terrestrial planets from the stage of isolated protoplanets. An outstanding issue concerning the conventional coagulation scenario for terrestrial planet formation is the natural expectation that cohesive collisions generally lead to modest eccentricity in the merger products (Chambers 2001). Although some damping mechanisms have been proposed (Ward 1989, 1993; Artymowicz 1993; Agnor & Ward 2002), these processes can also prevent the orbital instability from promoting growth (Iwasaki et al. 2001, 2002). Since protoplanetary growth beyond isolation requires orbit crossing, which tends to commence only after the damping effect has become very weak, the final products may still be eccentric (Kominami & Ida 2002, 2004).

In this paper we have proposed a mechanism through which the eccentricity growth is excited through a sweeping secular resonance (Ward et al. 1976) rather than dynamical instability among the embryos. In contrast, eccentricity damping is primarily due to the embryos' tidal interaction with the disk. With this separation between the excitation and damping processes, the embryos can attain equilibrium eccentricity while they are locked onto the sweeping secular resonance. Collisions can dislodge the trapped embryos from the resonance and their eccentricity may then be damped. The advantages of the idea of the sweeping secular resonance plus tidal damping are as follows:

1. Orbital disturbance is caused throughout almost all of the terrestrial planet region, and the efficiency of this process does not depend on embryos' mass.

2. The emerging protoplanets are circularized by strong damping since the accretion occurs in the time when tidal drag is still effective.

The core of Jupiter (as well as those of the terrestrial protoplanets) is formed before disk depletion and the capture of Jupiter's gas envelope ensue. Disk depletion is begun after the completion of Jupiter (§ 2). The disk applies tidal torque to the protoplanets and stops the enhancement of their eccentricities (§ 2.2). The presence of the fully formed Jupiter then gives rise to a secular resonance inside the solar system (§ 3).

The effect of the resonance is twofold: one effect is to trigger the migration of protoplanets, and the other is to stir the system at the time of the resonance. The former is stronger for embryos with larger mass, and the latter is more effective for low-mass embryos.

Near resonance, the eccentricities of the embryos oscillate with large amplitudes due to the exchange of angular momentum with Jupiter. Since the mass of Jupiter is far larger than that of the protoplanets, they unilaterally receive large eccentricity. Whenever planetary eccentricity increases, the planet receives strong tidal drag, and it moves toward the central star as required by the conservation of energy. This timescale is $\sim 250(m/M_{\oplus})^{-1}(a/1 \text{ AU})/e^2 \text{ yr}$ in a minimum-mass nebula. Because the forced eccentricity does not depend on the mass of protoplanets, the bigger planets are affected most strongly by the damping. The migration stops as the eccentricity is damped. The eccentricity is nearly constant during the migration.

As the disk depletes, the secular resonance moves inward on several times the depletion timescale. When protoplanetary migration speed is faster than that of the resonance, the resonance lags behind the protoplanet. When the migration is slow, resonance passes the protoplanet by. At this time, the eccentricity of the protoplanet is highly pumped up. Although the excitation magnitude caused by the sweeping secular resonance is independent of the mass, the smaller planets attain higher eccentricities and collide more readily with other bodies, since they experience weaker damping. The migration rate of bodies captured by the sweeping secular resonance does not depend on the mass of the protoplanet, unlike the migration caused by the forced eccentricity. Larger bodies receive strong drag due to their mass, and the smaller ones receive strong drag because of their high eccentricity. The numerical simulations of nonmutually gravitating protoplanets show that migration with the secular resonance almost always occurs within the parameters of the present solar system. Protoplanets suffering collisions or close encounters with their neighbors can separate from the sweeping secular resonance in a later stage. When the planets leave the resonance, their eccentricities are damped and those protoplanets with relatively large masses and orbital separations become stable for a while. However, whenever the secular resonance passes through one of the bodies further inward, the system is shaken again and growth of planets is promoted.

Due to the secular perturbation of Saturn on Jupiter, the ν_5 resonance is stalled at 0.6 AU. The embryos interior to 0.5 AU are not directly affected by the sweeping secular resonance. Their growth beyond isolation mass requires dynamical instability. Consequently, their orbits are likely to have a modest eccentricity, as is the case for Mercury.

The strength of the sweeping secular resonance is determined by the power of the Jovian planets (the eccentricity and mass)

and the speed of the sweeping (width of the system/disk depletion timescale). The strength of tidal damping is dependent on the masses of the embryos. The present Jovian eccentricity and the disk depletion timescale indicated by observations are sufficient to cause isolation-mass embryos in a minimum-mass nebula to undergo orbit crossing. However, the orbit of Jupiter and Saturn may have undergone orbital evolution. For example, Saturn may have been closer to Jupiter than it is today (Hahn & Malhotra 1999; Tsiganis et al. 2005). In this case, the propagation of the ν_5 resonance may be stalled at a location farther outside the orbit of Venus, in which case the terrestrial planets may be formed with more expanded orbits. But it is not clear whether the orbits of these planets would remain stable as ν_5 continues to propagate inward. We will consider the possibilities of a range of disk masses and gas giant orbital properties in the future.

Through the hypothetical sequence of events including the dynamical shake-up, the following properties of solar system planets can be accounted for by the sequential accretion scenario (Lin et al. 2005):

1. The high concentration of heavy elements in the envelopes and the possible presence of several Earth-mass cores in Jupiter and Saturn (Wurchterl et al. 2000; Guillot et al. 2004) provided the original motivation (Stevenson 1982a) and a conceptual basis for the assumption that the initial formation of their cores is followed by gas accretion.

2. The absence of major planets in the asteroid region has been interpreted as evidence for their growth suppression due to the perturbation induced by a rapidly formed Jupiter (Wetherill 1989). The present-day low surface density of minor planets suggests that most residual grains, planetesimals, and protoplanetary embryos were cleared out of this region.

3. The age dispersion between the calcium-aluminum-rich inclusion and ordinary chondrules within the same meteorites suggests that the epoch of planetesimal formation persisted for $\sim 2\text{--}3 \text{ Myr}$ (Amelin et al. 2002).

4. The observational signatures for disks around young stars in nearby clusters appear to decline on a timescale of $3\text{--}10 \text{ Myr}$ (Haisch et al. 2001). This decline is probably associated with gas depletion and grain growth (D'Alessio et al. 2001; Shuping et al. 2003; McCabe et al. 2003; Carpenter et al. 2005). With a modest concentration of heavy elements near the snow line, it is possible for the formation of multi-Earth-mass ($M_{\oplus}$) cores and gas giant planets to take place prior to severe depletion of the disk (Ida & Lin 2004).

5. The difference between the abundance of radioactive hafnium in the primitive meteorites and that of the Earth's mantle has been interpreted to suggest that the Earth differentiated while it acquired most of its mass on a timescale $< 10\text{--}30 \text{ Myr}$ (Yin et al. 2002; Jacobsen 2005). In contrast to the gas depletion timescale of the disks, the final assemblage of the terrestrial planets appears to have occurred after the emergence of Jupiter (also see E. W. Thommes et al. 2005, in preparation).

6. The compositional similarity between the Moon and the Earth upper mantle has been interpreted as evidence that the former was formed through collision between the latter and a Mars-size embryo after the embryos, and possibly the Earth, had differentiated (Benz et al. 1986; Canup & Asphaug 2001).

Finally, the eccentricity distribution of known extrasolar planets appears to be uniform, ranging from 0 to 0.7. Many

extrasolar planets also appear to have larger masses and shorter periods. In these systems, the sweeping secular resonance may have even greater impact in the dynamical shake-up of any residual planetesimals. We plan to extend our investigation to more general planetary configuration in the future.

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